Project Report

On

FLAN-T5 AND GEMINI-BASED VIDEO SUMMARIZER

"Revolutionizing Content with Context-Aware Summaries"

Submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY AND ENGINEERING

by

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June 2025

CERTIFICATE

This is to certify that the project report entitled **"FLAN-T5 AND GEMINI-BASED VIDEO SUMMARIZER"** submitted by **Manupati Jaideep (2021BITE083)** and **Kodi Vidya Sagar (2021BITE091)** to the Department of Information Technology and Engineering, National Institute of Technology Srinagar, in partial fulfillment of the requirements for the award of the degree of B. Tech in Information Technology and Engineering, is a Bonafide record of project work carried out by them under my supervision.

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Date: 01 July 2025

Place: Srinagar

STUDENT'S DECLARATION

We declare that this project report titled **"FLAN-T5 AND GEMINI-BASED VIDEO SUMMARIZER"** submitted in partial fulfillment of the requirements for the award of the degree of B. Tech in Information Technology and Engineering is a Bonafide record of project work carried out by us under the supervision of Dr. Prabal Verma.

The matter embodied in this project, in full or in parts, has not been submitted to any other Institution or University for the award of any degree or diploma. We also declare that the work submitted by us is entirely original, free from plagiarism, and has been diligently checked to ensure its authenticity.

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ABSTRACT

The rapid advancement of Natural Language Processing (NLP) has enabled the development of text summarization systems, yet video summarization—particularly for educational and multilingual content—remains underexplored. Existing models such as mT5 with NLLB and standalone FLAN-T5 provide summarization capabilities but suffer from key limitations including lack of difficulty control, insufficient multilingual support, and absence of interactive feedback. Most tools generate rigid or overly generic summaries and often ignore audio accessibility, which restricts usability in diverse learning environments. To overcome these challenges, we propose a robust, modular video summarization system that integrates several cutting-edge components. Our pipeline begins with transcript extraction using yt-dlp and Faster Whisper, followed by language detection and translation through Google Translator for supporting languages like Hindi. The cleaned transcript is processed in chunks by FLAN-T5, and the final summary is synthesized using Gemini 2.5 Pro, conditioned on user-selected difficulty levels: easy, medium, or hard. The summarized text is further feedback. Quantitative evaluation using ROUGE, BLEU, and BERT Score reveals significant improvements over baseline models, with ROUGE-1 reaching 0.569 and BERT Score peaking at 0.915. These results, along with qualitative feedback, demonstrate that our system effectively addresses the limitations of prior models by providing personalized, multilingual, and interactive summarization—marking a significant contribution to the domain of AI-powered educational tools converted into an audio podcast using gTTS, enhancing accessibility for auditory learners. Additionally, we embed a responsive Q&A interface powered by LLaMA3.2 via Ollama, enabling real-time chatbot interactions over the summary context. The entire architecture is deployed using a FastAPI backend and a React frontend, offering a seamless interface for uploading video files or URLs, viewing summaries, listening to podcasts, and submitting.

Key Words: Video Summarization, Multilingual Support, Multimodal Integration, Large Language Models, Personalized Content Summaries

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List of Symbols & Abbreviations

1. **AI** – Artificial Intelligence
2. **Q&A** – Question and Answer
3. **ASR** – Automatic Speech Recognition
4. **ROUGE** – Recall-Oriented Understudy for Gisting Evaluation
5. **BLEU** – Bilingual Evaluation Understudy
6. **BERTScore-F1** – Balanced Recall and Precision F1 Score
7. **NLP** – Natural Language Processing
8. **LLM** – Large Language Model
9. **YT-DLP** – YouTube Dynamic Link Processor
10. **FFMPEG** – Fast Forward Moving Picture Experts Group
11. **FLAN-T5** – Fine-tuned Language Net – Text-to-Text Transfer Transformer
12. **GEMINI 2.5 PRO** – Gemini 2.5 Professional Language Model
13. **Faster-Whisper** – Faster Whisper Automatic Speech Recognition System
14. **Deep Translator** – Deep Translation Tool
15. **API** – Application Programming Interface
16. **IoT** – Internet of Things
17. **NLLB** – No Language Left Behind
18. **CLIP** – Contrastive Language–Image Pretraining

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The sudden and explosive rise in video content, particularly in the education industry, has resulted in an unstoppable appetite for sophisticated devices to chop up long and convoluted content into bite-sized and absorbable summarizations. Educational videos, with tightly condensed information in them, need creative solutions to enable users to explore and understand their essential takeaways without having to play the entire content. Summarization of videos thus becomes an equally crucial task in the domains of Multimodal Artificial Intelligence (AI) and Natural Language Processing (NLP). The fundamental aim of video summarization is to reduce the cognitive burden of users by pulling out the most important information from very large video content in a manner that is both meaningful and understandable.

In spite of the existence of available summarization tools, there is still a big loophole in their capacity to address the diverse and changing demands of users. For example, in the context of multilingual educational material, including lectures conducted in Hindi or local languages other than English, most tools are incapable of generating precise or contextually specific summaries. In addition, the lack of customization options to fine-tune summaries according to different levels of user comprehension – for example, beginners, intermediates, or experienced learners – also reflects another limitation in today's technology. In the same vein, these tools' limitation in providing interactive question-answer interfaces reduces their efficacy in promoting deeper understanding and user engagement with the content.

In response to these limitations and filling gaps in meeting learners' and educators' needs, this research endeavor is concerned with the creation of an AI-driven video summarization tool. The envisioned system is meant to address the shortcomings of existing solutions by presenting core features like multilingual summarization, the possibility of varying

complexity levels of summaries, and an interactive question-and-answer interface facilitating active learning. These features are brought about through the use of cutting-edge AI models such as Flan-T5 and Gemini 2.5 Pro, which perform exceptionally well in creating high-quality, coherent summaries for various forms of content and languages. Furthermore, the use of local Large Language Models (LLMs) allows the tool to work offline with context-aware question-answering capabilities to aid in better understanding.

The suggested solution is notable in its focus on 2uture2y2lityy, 2uture2yy, and flexibility. With its provision for multilingual summarization, the tool makes it possible to include users of all linguistic backgrounds. The tool's ability to provide summaries at various levels of complexity makes it a valued resource for students at different levels of their educational pursuit. In addition, the interactive question-answering capability closes the disparity between passive consumption and active learning and promotes greater retention and understanding of information.

Essentially, this new video summarization technology is imaged as a revolutionizing tool for students, teachers, and assistive learning systems. Through the filling of key shortcomings in current tools and providing a very flexible and interactive solution, this research strives to revamp how video content is being watched, making it more effective, interactive, and for everybody. By integrating the latest AI technologies and an end-user focus, this system aims to transform the learning environment and serve the heterogeneous learning requirements of a global community.

What is Video Summarization?

Video summarization refers to the automatic creation of an abridged version of a video highlighting its most crucial or vital moments. This requires a review of the content of the video and the picking of significant scenes, frames, or sequences that encapsulate the primary concept or narrative, while omitting less important details. Summarization may be done in many forms, including:

1. **Static Summaries**: Symbolized by a series of key frames.

2. **Dynamic Summaries**: Embodied in brief video segments.
3. **Multimodal Summaries**: A combination of audio, text, and visual.

Why is Video Summarization Beneficial?

1. **Time-Efficient Content Consumption**:

    - Decreases the time needed to comprehend the central theme of long videos.
    - Helpful for audience members who prefer instant insights, such as in learning or business environments.

2. **Cognitive Ease:**

    - Summaries alleviate the cognitive load of deciphering large quantities of video information.
    - Emphasizes on key information, allowing complex content to be better understood.

3. **Accessibility**:

    - Facilitates users with language difficulties, hearing loss, or inadequate technical skills by providing subtitles, translation, or visual summaries.

4. **Content Search and Discovery**:

    - Allows rapid browsing and indexing of videos for big archives or streaming sites.
    - Allows content curation for particular user interests or contexts.

5. **Enhanced Learning**:

    - Adapts content to learners' levels of comprehension, improving personalized learning.

Applications of Video Summarization

1. **Education and E-Learning**:

    - Summarizing long lectures, tutorials, or webinars for rapid revision.
    - Content reshaping for varying age groups or levels of expertise.

2. **Social Media and Streaming Platforms**:

    - Trimming or highlights creation from films, shows, or sporting events.
    - Recap of user-generated content for greater engagement.

3. **Enterprise and Business**:

    - Executive summary of meetings, presentations, or training sessions.
    - Effective video conferencing summaries and findings for decision-makers.

4. **News and Media**:

    - Pulling out significant points from interviews, breaking news, or press conferences.
    - Condensing multimedia content for news aggregation sites.

5. **Security and Surveillance**:

    - Summarizing surveillance footage to determine critical incidents as soon as possible.
    - Helping forensic analysis by pointing out major events.

6. **Healthcare**:

    - Summarizing medical consultations or procedures for patient education.
    - Simplifying examination of surgical or diagnostic video recordings.

7. **Sports and Entertainment**:

- Emphasizing critical game moments, like goals or milestones.
- Condensing event coverage for audiences who have missed live streams.

8. **Personal Use**:

- Trimming personal videos to brief summations, like travel logs or celebrations.
- Organizing and storing collections of personal videos.

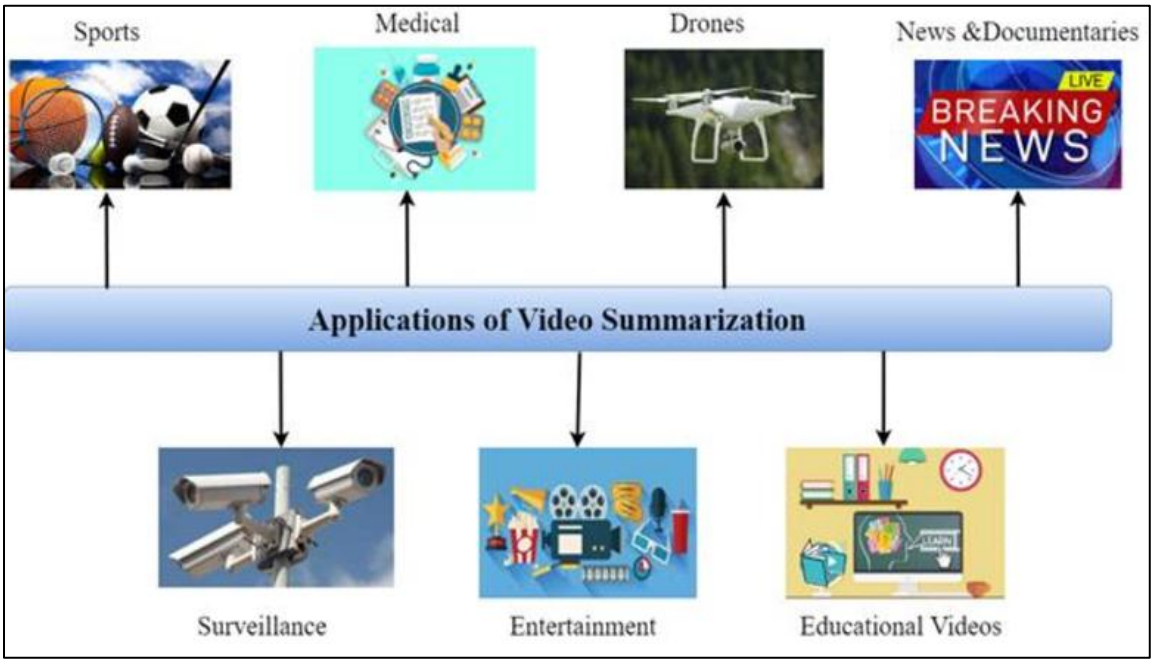


Figure 1 Applications of Video Summarization

1.2 Motivation

With the rapid pace of today's learning environment, students are often bombarded with long video content that has the potential to be overwhelming, particularly if they are short on time or dealing with complicated materials. Educational videos are great sources of knowledge, but their length and depth can create obstacles to effective learning and understanding. Current video summarization software does struggle to resolve this issue but is lacking in a number of important areas. Most significantly, these tools do not have the capacity to handle multilingual content efficiently, thus rendering non-English speakers

and users interacting with regional languages inadequately served. In addition, they do not provide personalized summaries that can be adapted to differing levels of experience, including beginner, intermediate, or advanced users. This uniform-fit model restricts the usefulness of these tools across different users.

Another key void in existing solutions is the lack of interactive engagement capabilities, including question-and-answer functions, that can facilitate increased comprehension of the material. These shortcomings make it imperative to create cutting-edge tools beyond mere content reduction that offer insightful and contextually aware information addressing the specific requirements of users.

The inspiration behind this research comes from an aspiration to fill these gaps by designing an accessible, efficient, and adaptable tool that can cater to various learning requirements. Through the convergence of multilingual support, personalization of summarization, and interactive engagement features, the research seeks to redefine the consumption of educational videos. It seeks to make learning more inclusive, efficient, and engaging for students, teachers, and lifelong learners worldwide. This effort not only solves the current limitation of existing efforts but also paves the way 6uturee developments in educational technology to make learning easier and more rewarding for learners across all segments.

1.3 Problem Statement

Current solutions lack the capability of offering multilingual, interactive, offline-compatible, personalized video summarization services. The current research focuses on creating a system that effectively solves these problems, thereby allowing customized, high-quality summaries and question-and-answer interaction in real-time for enhancing usability and accessibility.

1.4 Objectives

1. **Develop a Multilingual Summarization Tool:** Create a system supporting Hindi and English, leveraging speech-to-text, NLP, and multimodal summarization for accurate, user-tailored video summaries.
2. **Enable Adjustable Summary Complexity:** Introduce categorization of content into easy, intermediate, and hard levels, allowing users to select summaries based on their comprehension needs.
3. **Incorporate Interactive Question and Answer:** Integrate real-time question-answering functionality using local LLMs to facilitate deeper understanding.
4. **Enhance User Personalization and Feedback:** Implement mechanisms to personalize summaries based on user preferences and interactions, supported by a feedback system for continuous improvement.
5. **Optimize Scalability and Real-Time Processing:** Ensure the system handles large datasets efficiently, delivering fast and scalable summarization for diverse video content.

CHAPTER 2

LITERATURE SURVEY

2.1 Contributions, Limitations, and Applicability of Seminal Studies

The growing need for effective ways to consume video content has prompted significant research into video summarization using cutting-edge AI technologies. Below is an analysis of the three seminal studies that form the foundation for this project:

1. **"Deep Learning for Video Summarization: A Survey" by Zhu and Li (2020)**
    - **Contributions:**
        - Comprehensive classification of deep learning techniques into reinforcement learning, supervised learning, and unsupervised learning.
        - Demonstrates the effectiveness of neural networks in keyframe extraction and understanding video semantics.
    - **Limitations:**
        - Does not integrate textual and visual data for multimodal summarization.
        - Lacks user-specific features like complexity classification or personalization.
2. **"Multimodal Deep Learning for Video Summarization" by Chen and Wang (2019)**
    - **Contributions:**
        - Combines textual and visual data using CNNs for visual features and RNNs for textual data.
        - Improves summary quality compared to traditional methods.
    - **Limitations:**

- Does not include complexity-based categorization or user-specific customization.
- Limited scalability and lacks real-time processing capabilities for large-scale applications.

3. **"BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" by Devlin et al. (2019)**
    - **Contributions:**
        - Provides deep contextual understanding for tasks like text summarization.
        - Captures nuanced relationships between text sequences through bidirectional processing.
    - **Limitations:**
        - Only supports text-based tasks, excluding visual or multimodal data integration.

2.2 Research Gaps

The reviewed studies highlight several key limitations in current video summarization approaches:

1. **Complexity Analysis:**
    - Existing methods lack mechanisms to classify content into different complexity levels (easy, intermediate, or hard).
2. **Personalization:**
    - Limited focus on tailoring summaries to individual user preferences or feedback.
3. **Real-Time Processing:**

- Few systems address the need for scalable, real-time summarization, essential for handling large datasets.

4. **Comprehensive Multimodal Integration:**

- Current approaches fail to fully integrate text, visuals, and context, missing opportunities to enhance user engagement.

2.3 Comparative Analysis of Existing Models and Proposed Model

Related Work	Application Domain	Methodology	Output Format	Limitations	Future Work Suggested
Deep Learning for Video Summarization Zhu & Li (2020)	General video summarization	CNNs, RNNs, Reinforcement Learning	Keyframes, visual summary	No models processing; lacks adaptive control	Suggested hybrid multimodal techniques
Multimodal Deep Learning for Video Summarization Chen & Wang (2019)	Visual-textual content (movies, talks)	CNN (visual), RNN (text), attention fusion	Combined video + text	No real-time support; lacks complex personaliz-ation	Expand multimodal processing, enable real-time
BERT: Devlin et al. (2019)	Text summarization, NLP pipelines	BERT Transformer (unsupervised pretraining)	Text summary	Not applicable to video or audio	Suggested downstream fine-tuning

GPT-family (OpenAI, etc.)	Generic text generation & summarization	Transformer LLMs (pretrained, generative)	Abstractive text summary	Not suitable for transcript-based summariz-ation	Multimodal pretraining recommender
Proposed Model	Educational lectures, academic content, public talks, podcasts, explainers, news, multilingual videos	Faster Whisper, Gemini 2.5, FLAN-T5, gTTS, Ollama with LlaMA3.2 (chatbot)	Markdown summary	-	Extendable to fine-tuned summaries and richer voices

Table 1 Comparative Analysis of Existing Models and Proposed Model

2.4 Relevance to Proposed Work

Building on the strengths of these studies, this project addresses the identified gaps by:

1. Implementing a complexity classification module to tailor summaries to varying user expertise levels.
2. Incorporating personalization features through a feedback-driven system to adapt to user preferences dynamically.
3. Leveraging multimodal summarization, combining text, visuals, and contextual understanding for richer summaries.

4. Ensuring real-time processing for scalable and efficient handling of large video datasets.

By addressing these challenges, this project hopes to reimagine video summarization, making it more user-centric, context-aware, and accessible, with applications ranging from education to training to productivity.

CHAPTER 3

METHODOLOGY

3.1 Components of Proposed Frame Work

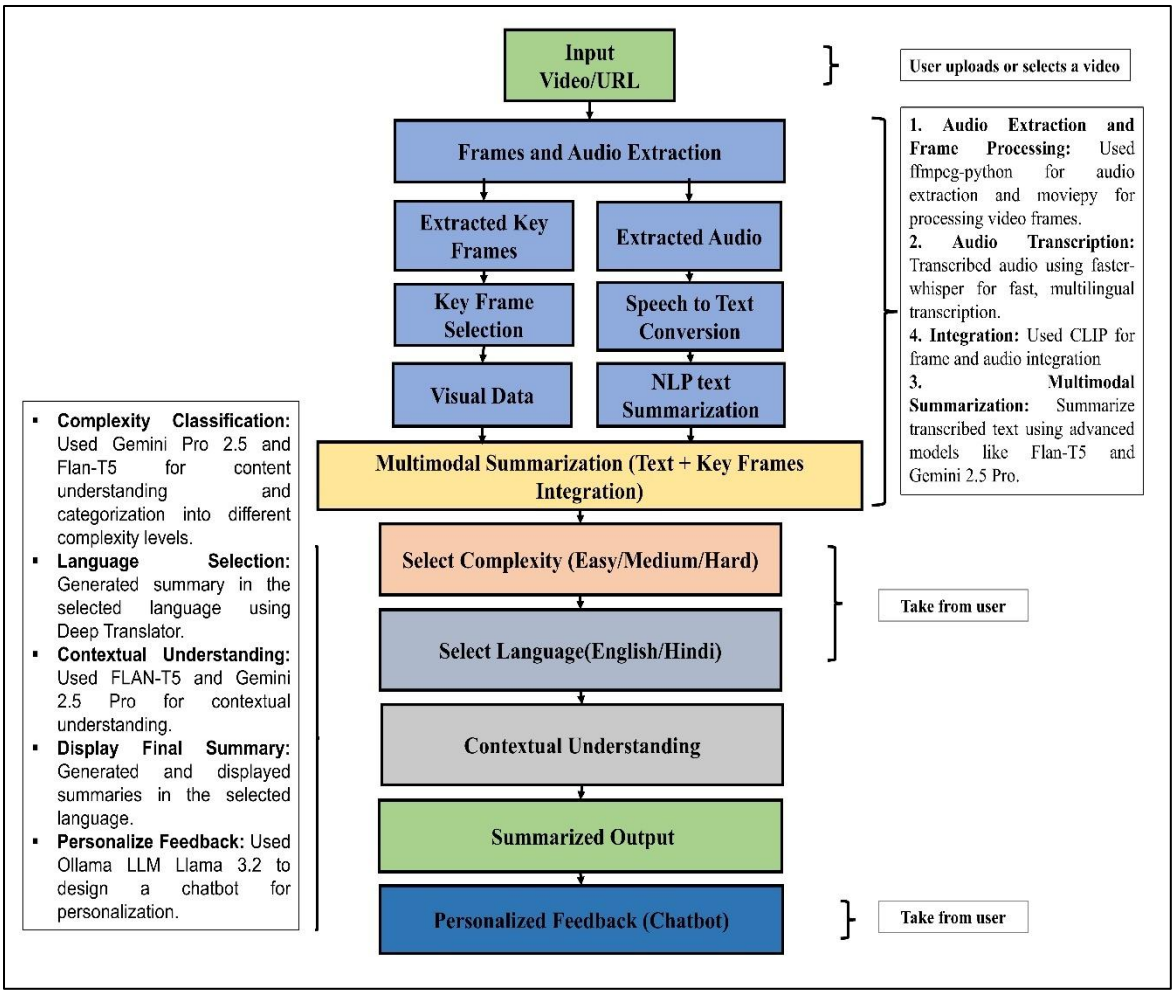


Figure 2 Components of Proposed Frame Work

The pipeline outlines the workflow of an AI-powered video summarization tool designed to extract, summarize, and personalize content from videos. The process begins with the user uploading a video or providing a URL as input, which is then subjected to frames and audio extraction. Tools like ffmpeg-python separate the audio and process video frames, while key frames are identified to represent the video visually. The audio is converted into text using faster-whisper, enabling multilingual transcription. This transcribed text

undergoes NLP-based summarization using advanced AI models like Flan-T5 and Gemini 2.5 Pro, producing concise summaries. These textual summaries are then integrated with visual data, such as key frames, using tools like CLIP, creating a multimodal representation for enhanced comprehension.

To personalize the output, users can select the complexity level of the summary (Easy, Medium, or Hard) and the preferred language (English or Hindi), facilitated by tools like Deep Translator. Advanced models like Flan-T5 ensure the summaries are contextually accurate. Once tailored inputs are provided, the system generates a high-quality, user-specific summary that combines text and visual elements. The process concludes with a personalized feedback mechanism powered by a chatbot designed using llama 3.2, allowing users to ask questions and interact with the summarized content for deeper understanding. This pipeline leverages cutting-edge technologies to deliver accessible, efficient, and adaptable summarization, meeting diverse user needs in educational and other domains.

3.2 Features of Proposed Model:

1. **Multimodal Summarization**: Integrates text, audio, and visual information for informative, contextually precise summaries.
2. **Multilingual Support**: Summarizes video content in various languages, such as Hindi and English.
3. **Customizable Complexity Levels**: Produces summaries suited to different user understanding levels—easy, intermediate, and advanced.
4. **Real-Time Processing**: Provides fast video/audio processing with low latency.
5. **Interactive Question and Answer**: Allows users to pose questions and get relevant answers from the summarized material through a chatbot.
6. **Offline Compatibility**: Runs offline with software such as Ollama and Faster-Whisper, allowing for functionality without dependence on internet connectivity.
7. **Evaluation and Feedback Loop**: Continually enhances the quality of summaries through feedback from users as well as through automatic validation measures such as ROUGE and BLEU.

8. **Efficient Transcription**: Leverages refined use of Faster-Whisper for efficient and rapid speech-to-text conversion.
9. **Seamless Translation**: Translates transcripts across languages for increased accessibility with Deep Translator.
10. **Dynamic Key Frame Selection**: Retrieves and combines visually important frames for richer multimedia summaries.
11. **Scalability**: Processes large sets of data effectively, making possible processing of bulk video content.
12. **Personalization**: Adjusts summaries based on individual users' preferences and lets users set preferences.
13. **Pickle Library Integration**: Saves processing time by storing and reusing preprocessed data.
14. **Audio Podcast**: The model generates audio podcasts from summaries, enhancing accessibility for visually impaired users and audio learners.
15. **Friendly Feedback System**: Gathers ratings, comments, and corrections for model refinement iteratively.
16. **Secure and Private**: Offers local execution to maintain data security and privacy.
17. **Open-Source Tools**: Utilizes strong libraries such as FFmpeg, Faster-Whisper, and yt-dlp to carry out preprocessing and transcription efficiently.

3.3 Tools and Technologies

1. FLAN-T5

FLAN-T5 (Fine-tuned Language Net – Text-to-Text Transfer Transformer) is a state-of-the-art natural language processing model designed by Google. It is a variation of the widely used T5 (Text-to-Text Transfer Transformer) model but fine-tuned on instruction-based data. FLAN-T5 is specially designed to carry out diverse language tasks by understanding natural language instructions. Contrary to traditional models, FLAN-T5 is a master of generalization to unseen tasks and hence very versatile and flexible for a variety of tasks.

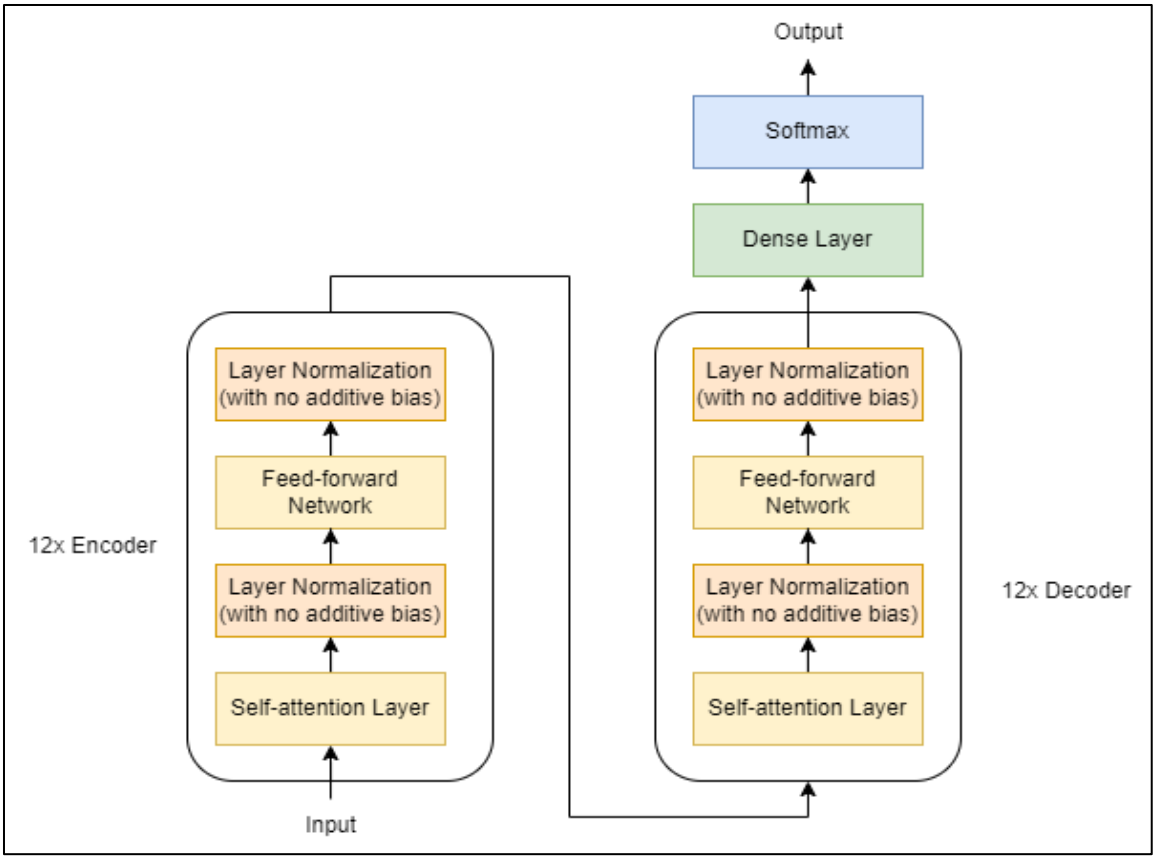


Figure 3 Architecture of FLAN-T5

Major Features of FLAN-T5:

1. **Instruction Tuning**:
    - FLAN-T5 is instructed on datasets to train the model to execute instructions well, which improves task comprehension.
2. **Text-to-Text Framework**:
    - It's run on an integrated framework in which every task, including summarization, translation, and question-answering, is framed as text-to-text problems for ease of workflow.
3. **Multilingual Capability**:

- The model can handle multiple languages, making it appropriate for applications with a variety of linguistic contexts.

4. **High Performance**:
    - FLAN-T5 achieves cutting-edge performance on various benchmarks, such as summarization, classification, and reasoning tasks.
5. **Adaptability**:
    - It shows remarkable performance in zero-shot and few-shot cases, which implies that it can perform tasks with very little task-specific fine-tuning.

Application of FLAN-T5 to the Project:

In this project, **FLAN-T5** is used to generate high-quality, abridged, and contextually correct video summaries. It is specifically employed for:

- **Contextual Understanding**: Understanding and summarizing the transcribed content so that the output is in accordance with the topic of the video.
- **Complexity Classification**: Classifying summaries into Easy, Medium, and Hard levels so that outputs can be optimized according to users' comprehension requirements.
- **Multimodal Summarization**: Working in conjunction with other models (e.g., Gemini 2.5 Pro) to give text summaries combined with visual data for greater user interaction.

Why FLAN-T5?

- FLAN-T5 was selected for its generalizability across tasks, its support for multiple languages, and its instruction-based tuning efficiency. These are the qualities that make it a reliable option for building an adaptable, accurate, and user-friendly video summarization system.

2. GEMINI 2.5 PRO

Gemini 2.5 Pro is a sophisticated AI model that is specifically tailored for complicated natural language processing (NLP) and multimodal applications as part of the Gemini family. It is highly proficient in managing operations involving the synthesis of text, audio, and visual information and hence is best suited for use in video summarization, sentiment analysis, and content creation. Gemini 2.5 Pro is specifically architected to provide high-quality output with enhanced accuracy and speed across multiple applications.

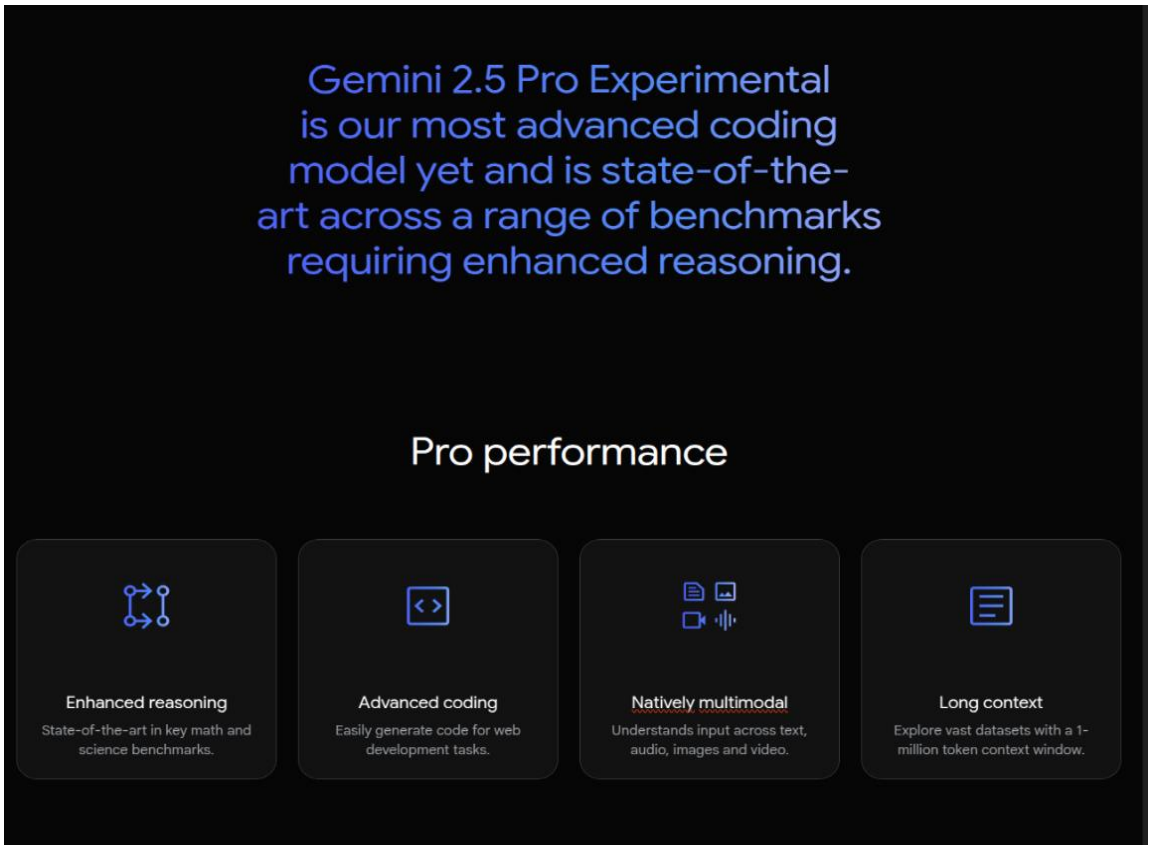


Figure 4 Feature of Gemini 2.5 Pro

Major Features of Gemini 2.5 Pro:

1. **Multimodal Integration:**
    - Facilitates the ease of processing and fusion of textual, auditory, and visual inputs, allowing for holistic understanding of multimedia data.

2. **Contextual Awareness:**
    - Is capable of interpreting and producing outputs in terms of the contextual implications of input information, providing more meaningful and precise results.
3. **Multilingual Capability:**
    - Provides strong multilingual support, making it perfect for summarization work in multiple languages.
4. **High Performance:**
    - Obtains the latest state-of-the-art performance on a range of benchmarks, providing even higher accuracy and relevance for structured and unstructured data.
5. **Scalability and Efficiency:**
    - Designed to process large datasets, enabling real-time computation for high-usage applications.

Use of Gemini 2.5 Pro in the Project:

In this project, Gemini 2.5 Pro is used for:

- **Multimodal Summarization:** Creating in-depth video summaries through the integration of transcribed text, key frames, and visual information.
- **Contextual Understanding:** Collaborating with models such as FLAN-T5 to provide contextually precise and content-related summaries.
- **Complexity Classification:** Assisting in the classification of content into Easy, Medium, and Hard levels to offer customized user experiences.

Why Gemini 2.5 Pro?

Gemini 2.5 Pro was selected for its high-quality multimodal processing capabilities, multilingualism, and effectiveness in working with large-scale data. They are the key

features that make it a critical part of the video summarization tool, providing complete, precise, and user-friendly summaries for a wide range of applications.

3. Faster Whisper

Faster-Whisper is a lightweight, high-performance version of OpenAI's Whisper model, which has been optimized for speech-to-text transcription operations. It gives a tremendous boost to speed and efficiency and is very much applicable for real-time and mass-scale applications. Faster-Whisper is most effective in multilingual transcription and can support almost any language, providing inclusivity and diversity.

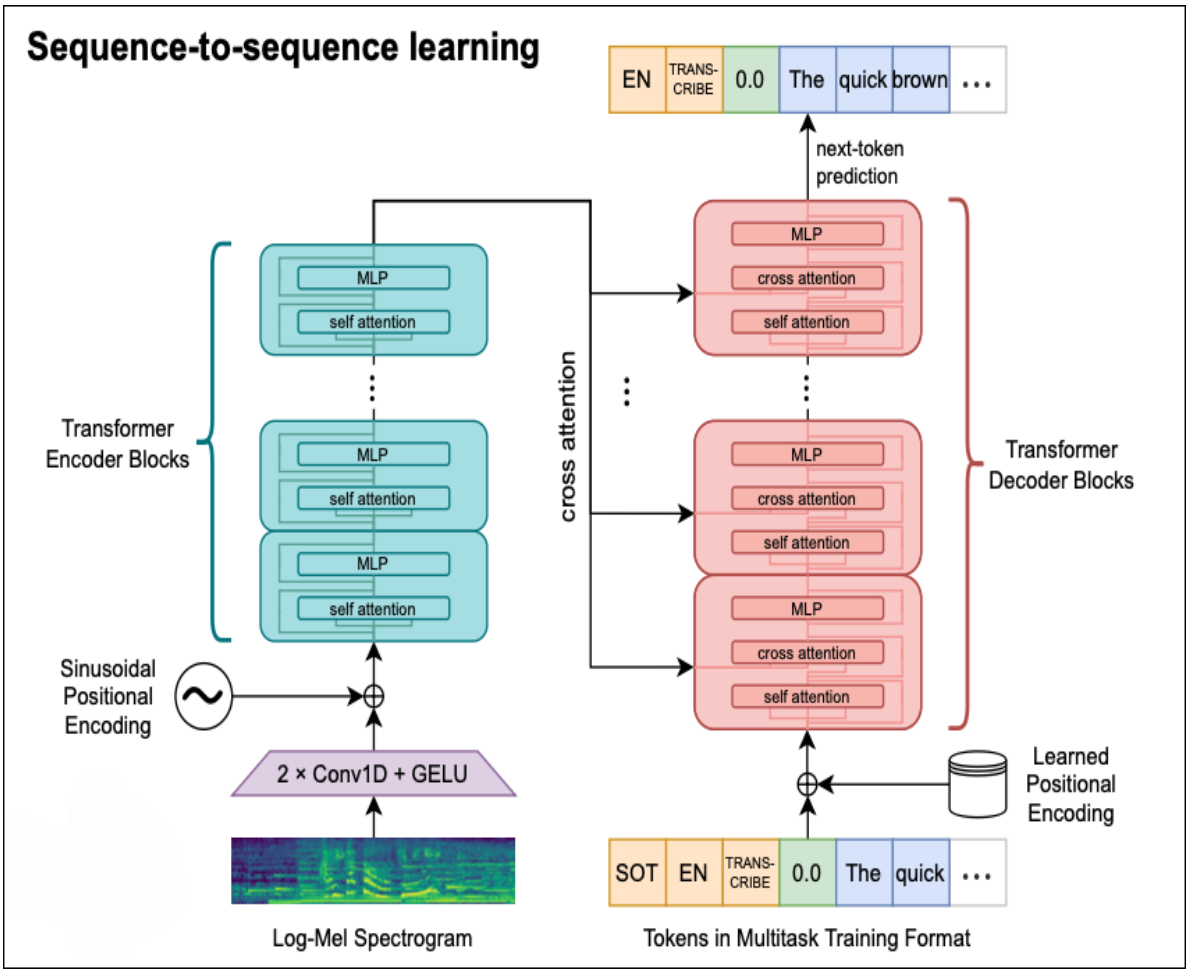


Figure 5 Architecture of Faster Whisper

Major Features of Faster-Whisper:

1. **Improved Speed and Efficiency**:
    - Optimized for quicker processing relative to the base Whisper model, making real-time transcription of videos and audio streams possible.
2. **Lightweight Architecture**:
    - Compact memory footprint ensures the model executes smoothly on hardware-constrained systems, including local settings.
3. **Multilingual Support**:
    - Can transcribe multiple languages of audio, which makes it highly suitable for global use cases and multilingual projects.
4. **High Accuracy**:
    - Preserves the Whisper model's capability to produce accurate transcriptions, regardless of difficult audio inputs, such as noisy environments.
5. **Scalability**:
    - Can effectively deal with large datasets, which makes it a good fit for projects that entail a lot of video or audio content.

Use of Faster-Whisper in the Project:

In the project, **Faster-Whisper** is used to:

- **Conversion of Speech to Text**: Transcribe video audio with high accuracy and support for multiple languages, thus allowing generation of text data for subsequent summarization.

- **Multilingual Support**: Process video in various languages, like Hindi and English, to ensure the tool supports a larger population.

- **Real-Time Processing**: Provide real-time transcription for videos to increase the efficiency of the summarization process.

Why Faster-Whisper?

Faster-Whisper was selected due to its high speed, high accuracy, and low latency. Its lightweight design makes it appropriate for local and resource-poor settings, and its multilingual support is exactly what the project hopes to achieve in terms of making video summarization inclusive and accessible.

4. CLIP (Contrastive Language–Image Pretraining)

CLIP is a revolutionary AI model designed by OpenAI that closes the gap between text and images in a single framework. Trained on an enormous number of text-image pairs, it can grasp and connect textual phrases with respective visual features accurately. CLIP brings visual and textual representations into a common embedding space by using contrastive learning, making it extremely flexible for various applications dealing with multimodal inputs.

Use of CLIP in the Project:

In the project, **CLIP** is applied to:

- **Multimodal Integration**: Integrate visual information (key frames) and text information (condensed transcriptions) to develop rich video summaries.

- **Key Frame Selection**: Locate and synchronize key video frames with the most applicable text segments for enhanced understanding.

- **Richer User Experience**: Present contextually correct visual and text summaries, allowing users to understand video content efficiently.

- **Why CLIP?**

  CLIP was selected due to its capacity to comprehensively merge textual and visual information, and it was a great match for multimodal summarization. Its zero-shot learning capacity and contextual sense guarantee that summaries produced are correct, interpretable, and user-friendly. Through the use of CLIP, the project gains more interactivity and usability, supporting various user needs.

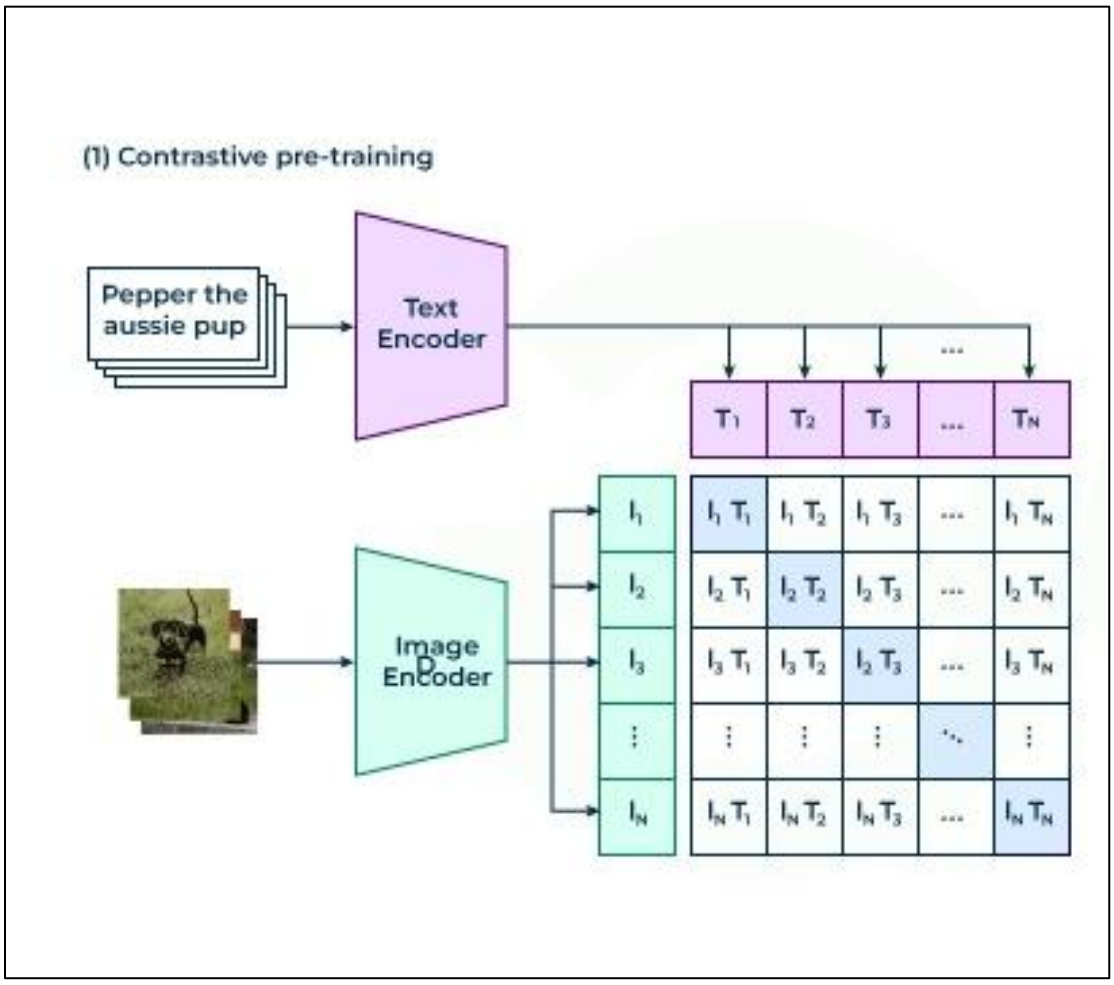


Figure 6 Architecture of CLIP_1

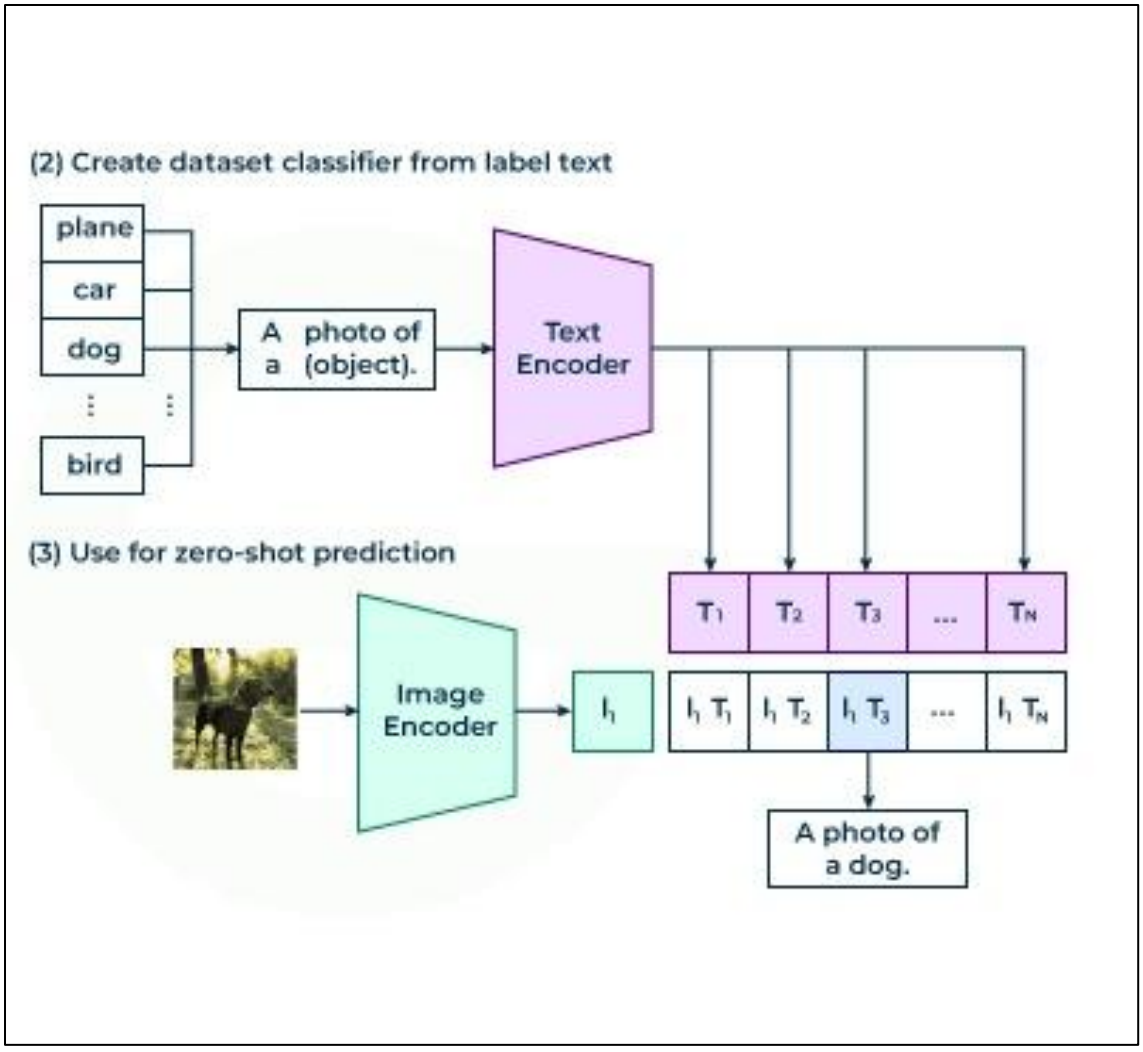


Figure 7 Architecture of CLIP_2

Major Features of CLIP:

1. **Multimodal Understanding**:
    - Handles text and images at the same time and develops a common semantic space for their representations.
2. **Zero-Shot Learning**:

- Performs tasks exceptionally well with no task-specific training, so it can generalize well to novel datasets and applications.

3. **Versatility Across Tasks**:

    - Facilitates operations such as image classification, object detection, captioning, and visual question answering through its capability to bridge text and image domains.

4. **Scalable Architecture**:

    - Based on a transformer architecture, enabling it to process and analyze rich data at large scales

5. **Contextual Awareness**:

    - Precisely links textual inputs to corresponding visual content, yielding meaningful and contextually correct outputs.

5. FFmpeg (Fast Forward MPEG)

FFmpeg is a universal, open-source multimedia framework that is commonly used for manipulating audio, video, and other types of multimedia files. It is a command-line program that offers robust libraries and facilities for a wide range of multimedia operations, including conversion, editing, compression, and streaming. FFmpeg is extremely efficient, cross-platform, and supports a vast array of formats and codecs, making it the first choice for developers and engineers developing multimedia applications.

Project Use of FFmpeg:

FFmpeg is used in the project for:

- **Audio and Video Extraction**: Extracting audio streams and important video frames from input files for analysis.

- **Preprocessing**: Prerequisites video and audio data so that it can be prepared for transcription and summarization.

- **Efficient Processing**: Processing large multimedia files with fewer resource requirements.

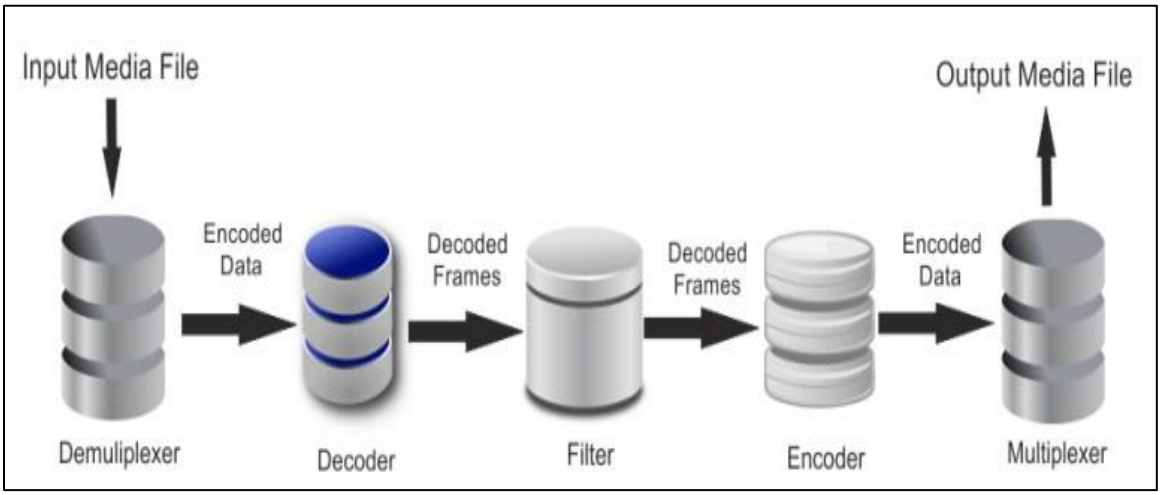


Figure 8 Working of FFmpeg

Major Features of FFmpeg:

1. **Wide Range of Format Support**:

    - FFmpeg contains support for nearly all multimedia formats such as MP4, AVI, MKV, MP3, AAC, and numerous others.

2. **Rich Set of Multimedia Tools**:

    - Provides tools such as ffmpeg (media processing), ffplay (media playback), and ffprobe (media inspection) for versatile multimedia operations.

3. **Lossless/Customized Conversion**:

    - Supports lossless or customized compression and conversion of audio and video formats.

4. **Frame and Audio Extraction**:
    - Separates individual frames from videos and extracts audio tracks for in-depth analysis or processing.
5. **Video and Audio Editing**:
    - Enables trimming, combining, cropping, filtering, and applying effects to audio and video material.
6. **Streaming Support**:
    - Supports real-time streaming and handles protocols such as HLS, RTMP, and MPEG-DASH.
7. **Cross-Platform Compatibility**:
    - Compatible with Linux, Windows, MacOS, and other operating systems, thereby being highly accessible.

Why FFmpeg?

FFmpeg was chosen due to its stability, performance, and feature-rich characteristics. Its capacity for performing diverse multimedia operations and accommodating almost all formats places it at the center of any project that includes video and audio processing. Its use in this project guarantees smooth extraction, preprocessing, and multimedia data management, allowing proper and efficient video summarization.

6. Ollama with llama3.2

Ollama is a high-end platform for running and engaging with local Large Language Models (LLMs) such as Llama 3.2 on personal devices. This configuration brings the power and flexibility of Ollama's infrastructure together with the cutting-edge capabilities of Llama

3.2, an advanced language model optimized for demanding natural language processing (NLP) applications.

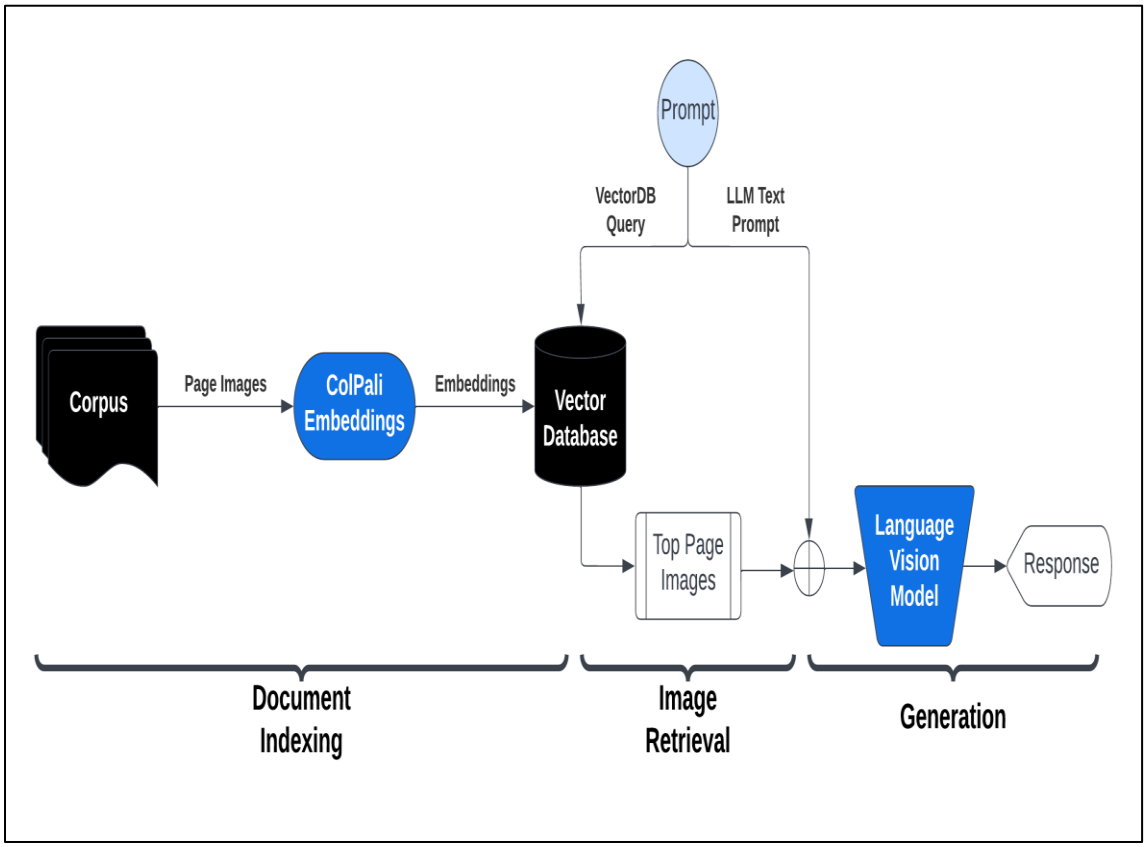


Figure 9 RAG With llama3.2

Major Features of Llama 3.2:

1. **Increased Efficiency**:
    - Optimized for speedier inference and reduced resource usage, thus fitting for on-premises deployments.
2. **Richer Contextual Comprehension**:
    - Can support long-context situations and produce accurate, contextual responses.

3. **Tuned Performance**:

    - Trained in a broad dataset and tuned for targeted tasks, guaranteeing high accuracy and relevance in use cases.

4. **Multimodal Support**:

    - Accommodates text, image, and even code-based interactions, thus being flexible to a myriad of use cases.

5. **Privacy and Security**:

    - Since it executes locally on Ollama, Llama 3.2 preserves user data as private, avoiding dependency on remote servers.

Why Use Ollama with Llama 3.2?

1. **Local Execution**:

    - Facilitates execution of sophisticated LLMs without needing cloud infrastructure, promoting privacy and data control.

2. **Customizability**:

    - Accommodates model fine-tuning to suit domain-specific needs or special applications.

3. **Resource Optimization**:

    - Natively integrated with personal or enterprise hardware, providing high efficiency on resource-scarce systems.

4. **Diverse Use Cases**:

- Perfect for chatbots, summarizing, question answering, code generation, and much more, with offline environment support.

Applications of Ollama with Llama 3.2:

1. **Interactive AI Agents**:
    - Developing local conversational agents for customer service or personal productivity.
2. **Document Summarization**:
    - Summarizing and processing high volumes of data locally for confidentiality and performance.
3. **Multimodal Analysis**:
    - Combining text and image information for use in education, design, and research.
4. **Enterprise Solutions**:
    - Creating secure, scalable AI systems for business process automation, data analysis, and internal tools.

Why Llama 3.2?

Llama 3.2 is the next generation in effective and precise NLP that offers the optimal blend of performance and scalability for local use. In combination with Ollama, it is a robust tool for creating secure, private, and highly customizable AI systems that can comply with current computational and ethical requirements.

3.4 Workflow of Proposed Model

The system operates through a pipeline that incorporates various tools and methodologies for video processing, transcription, translation, summarization, and evaluation. The pipeline is structured as follows:

1. **Input and Extraction**: Handles video/audio input and extracts necessary audio or frames.
2. **Transcription**: Converts extracted audio into text form.
3. **Translation**: Enables multilingual support by translating transcripts to the required language.
4. **Summarization**: Generates contextually relevant summaries tailored to different complexity levels.
5. **Interactive Q&A**: Provides a chatbot-based question-answering system for user engagement.
6. **Evaluation and Feedback**: Validates summary quality and collects user input for iterative improvements.

The proposed system works in a step-by-step process. Each stage is meticulously designed to ensure high accuracy, scalability, and adaptability for diverse video content and user needs. The methodology is divided into the following stages:

1. Video/Audio Input and Extraction

- **Objective**: Prepare video/audio files for transcription by extracting audio in a standardized format.
- **Input Sources**:
    - YouTube or Vimeo URLs.
    - Uploaded files in various formats, including MP4, MP3, WAV, and M4A.
- **Tools and Libraries Used**:

- yt-dlp: For downloading and extracting audio from video URLs.
- moviepy: For audio extraction from uploaded video files.
- ffmpeg-python: Underlying tool for format conversions.
- os, shutil, tempfile: Manage temporary file storage.

- **Process Flow**:
  1. If input is a URL: Download video and extract audio in WAV or MP3 format.
  2. If input is a file: Determine file type (audio or video) and extract audio if needed.
  3. Store extracted audio in a designated temporary directory (e.g., /temp_audio/).

2. Transcription

- **Objective**: Convert the extracted audio into text for further processing.
- **Tools and Libraries Used**:
  - faster-whisper: A fast and accurate multilingual automatic speech recognition (ASR) engine.
  - torch, numpy: Core dependencies for running ASR models.

- **Process Flow**:
  1. Load the extracted audio file (e.g., WAV or MP3).
  2. Use faster-whisper to transcribe the audio with timestamped segments if required.
  3. Output a plain-text transcript in the language of the original audio.

3. Translation

- **Objective**: Provide multilingual support by translating transcripts into the desired language for summarization or user display.
- **Tools and Libraries Used**:

- deep-translator with Google Translator backend.

- **Process Flow**:
  1. Detect the language of the transcript using built-in confidence scoring from faster-whisper.
  2. Translate non-English transcripts to English for summary generation (if required).
  3. Translate the final summary back into the original language if needed for user output.

4. Summarization

- **Objective**: Generate summaries that are concise, contextually relevant, and tailored to the user's comprehension level.
- **Tools and Libraries Used**:

  - Flan-T5: For lightweight summarization tasks.
  - Gemini 2.5 Pro: For advanced, high-quality, structured summaries.

- **Process Flow**:
  1. Use dynamic prompts to generate summaries with varying complexity levels (easy, intermediate, advanced).
  2. **Level-based summaries**:
     - **Level 1**: Simple bullet points.
     - **Level 3**: Instructional-style summaries.
     - **Level 5**: Conceptual and in-depth summaries.
  3. Combine extractive and abstractive summarization techniques to preserve context and key details.

  - **Parallelized Summarization with Thread Pool Executor**

    Purpose: Speeds up summary generation by processing multiple chunks in parallel.

Key Features:

- max_workers=4: Creates a thread pool with 4 workers.
- executor.map (): Distributes the workload, applying summarize_chunk to each chunk concurrently.
- list (): Collects the parallelized results into a list (summaries).

- **Advantages:**
  - Reduces processing time significantly compared to sequential execution.
  - Ideal for tasks involving external APIs that support concurrency.
- **Parallel Execution (Faster with Thread Pool Executor):**
  - Processes multiple chunks simultaneously, leveraging multi-threading for efficiency.

5. Interactive Question and Answer

- **Objective**: Enable users to interact with the summarized content through a chatbot, asking questions and receiving relevant answers.
- **Tools and Libraries Used**:
  - Ollama: Local deployment of LlaMA-based language models for offline Q&A.
  - Fast API: Backend framework to stream chatbot responses.
- **Process Flow**:
  1. Use the generated summary as context for the chatbot.
  2. Maintain session-based memory for multi-turn conversations.
  3. Stream responses to the frontend using a generator-based approach.

6. Evaluation and Feedback

The Evaluation and Feedback phase is an important element in enhancing the performance and adaptability of the suggested video summarization model. This phase confirms the

system provides high-quality, user-personalized summaries through validation of outputs and feedback from users to make iterative improvements. The process goes as follows:

1. Output Validation

- **Automatic Metrics**:
    - Metrics such as ROUGE, BLEU, or METEOR are employed to measure the generated summary against ground-truth summaries.
    - These metrics assess aspects such as coverage, coherence, and fluency of the summary.
- **Manual Validation**:
    - Users or the evaluators manually validate the summary for accuracy, completeness, and relevance with respect to the video content.
    - This step helps ensure that subjective features like readability or contextual relevance are given due attention.

2. User Feedback Collection

- **Feedback Forms**: Users can provide ratings for the summary on predetermined criteria like accuracy, brevity, and readability (e.g., a 5-star rating system).
- **Open-Ended Comments**: Users can offer more suggestions or indicate inaccuracies in the summary.
- **Interactive Corrections**: Users can highlight certain sections of the summary that need to be corrected or ask for a re-summary with varying settings (e.g., a simpler version).

3. Analysis of Feedback

- **Quantitative Analysis**: The ratings and metrics are tallied to determine overall trend performance in areas such as conciseness.

- **Qualitative Analysis**: Correct requests and comments are examined to determine user preference or frequently occurring issues.

4. Recurring Model Refinement

- **Retraining**: New data representing user preferences or error fixes are used to retrain the system.
- **Algorithm Updates**: Subsystems such as summary difficulty classification or question-answering sub-systems are tuned to meet user-specific requirements.
- **Parameter Optimization**: Model hyperparameters are tuned for performance on key metrics.

5. User Adaptation

- **Personalized Recommendations**: Customizing summaries based on historical user behavior and preferences.
- **Feedback-Based Configurations**: Permitting users to predefine preferences (e.g., "Always provide easy-level summaries").

6. Continuous Evaluation Loop

- Summaries are created and confirmed.
- Feedback is gathered and examined.
- Model updates are applied.
- The loop continues, refining performance incrementally.

3.5 Mechanism of Summarizer

Explanation:

1. Audio Transcription

- Input: Raw audio stream (16kHz, mono)
- Processing: Faster-Whisper with VAD (Voice Activity Detection)

- Output: Timestamped transcript with speaker diarization and confidence scores

2. Context-Aware Chunking

- **Segmentation Strategy:**
    - Primary splits at speaker changes
    - Secondary splits at semantic boundaries (BERT-based topic shift detection)
- **Adaptive Sizing:**
    - Base: 512 tokens for dense technical content
    - Extended: 1024 tokens for conversational segments
- **Overlap Management:**
    - 25% content overlap between adjacent chunks
    - Overlap regions marked with special context tokens

3. Thread Pool Execution

- **Architecture:**
    - Dynamic worker pool (size = 2 × available GPUs)
    - Priority queue sorted by chunk complexity score
- **Processing Workflow:**
    1. Master node dispatches chunks to idle workers
    2. Workers process chunks through dual-model NLP pipeline
    3. Results stream to integration buffer with completion flags
- **Optimizations:**
    - Zero-copy data transfer via CUDA memory
    - Work-stealing for load balancing

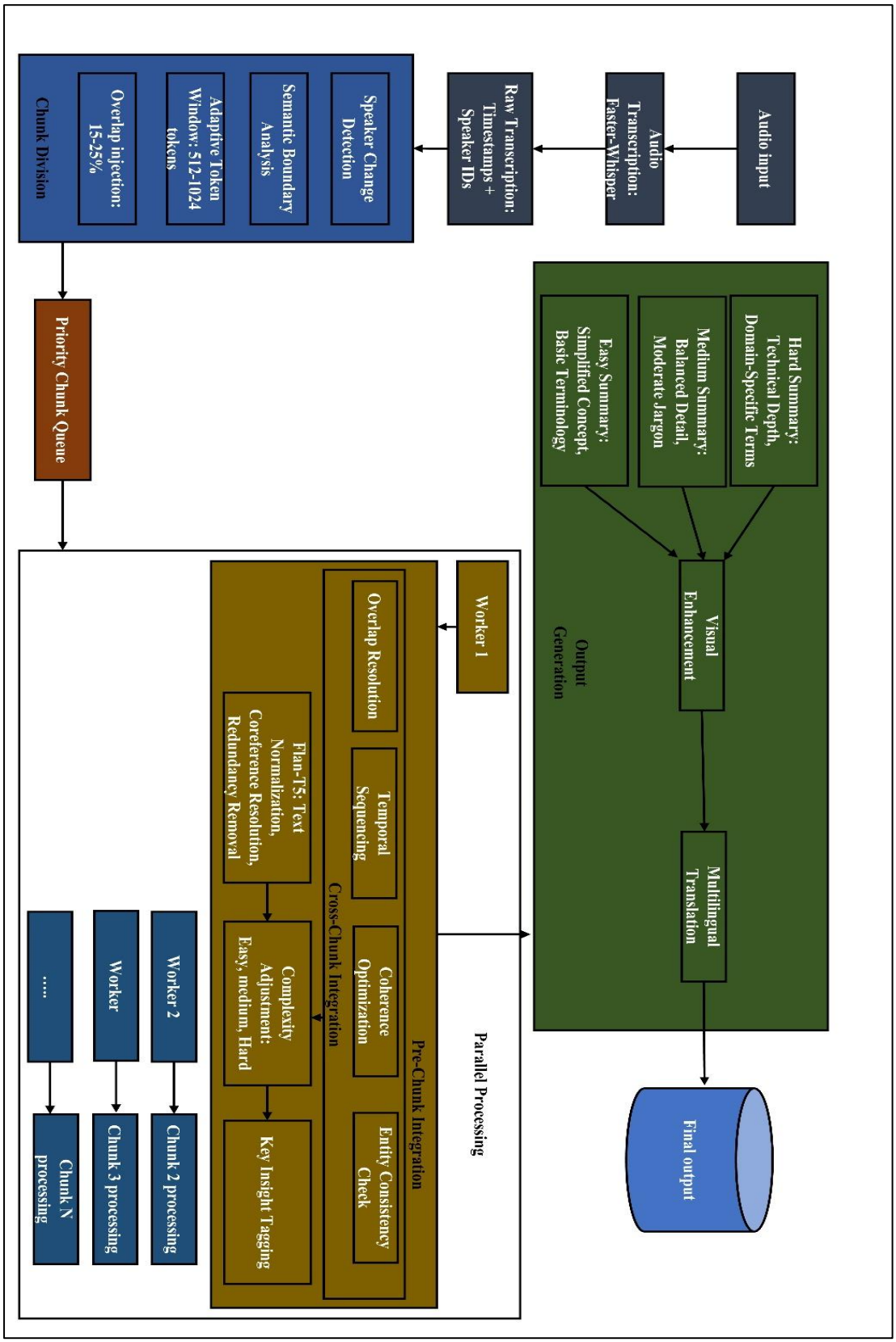


Figure 10 Summarization Mechanism

4. Dual-Model NLP Processing

- **Flan-T5 Specialization:**
    - Sentence compression and normalization
    - Pronoun resolution (coreference chains)
- **Gemini 2.5 Pro Specialization:**
    - Cross-chunk entity consistency
    - Technical term extraction and linking
    - Salience scoring (0-5 scale) using:
      Salience = 0.4 * Frequency + 0.3 * novelty + 0.3 * position

5. Complexity Adjustment

- **Level-Specific Processing:**

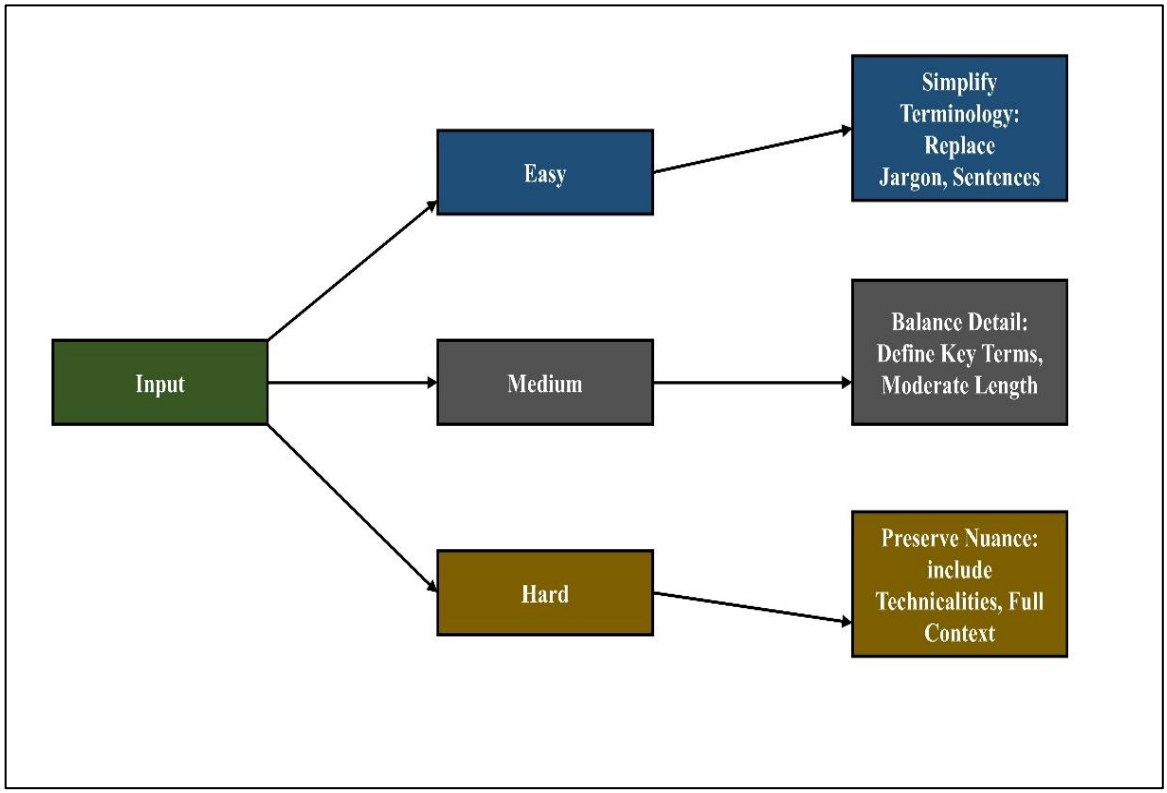


Figure 11 Level Specific Processing

- **Adaptive Techniques:**
  - Lexical simplification (Easy)
  - Controlled term expansion (Medium)
  - Domain-specific augmentation (Hard)

6. Cross-Chunk Integration

- **Key Operations**:
  - Overlap reconciliation using BERT-based similarity matching
  - Temporal sequencing with timestamp alignment
  - Entity consistency validation across chunks
  - Transition generation for narrative flow
- **Quality Assurance**:
  - Perplexity checks (<50 for fluency)
  - Entity recall verification (>90%)
  - Redundancy elimination (ROUGE-L < 0.3)

7. Multi-Level Output Generation

- **Summary Tracks**

Level	Target Audience	Content Features	Length Ratio
Easy	General Public	Simple analogies, no jargon	5-7% of original
Medium	Educated Readers	Balanced detail, some terms	10 – 12%
Hard	Domain Experts	Technical depth, full context	15-20%

Table 2 Multi Level Output Generation

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Evaluation Overview

The evaluation of the proposed video summarization model involves comparing its performance to existing models using various metrics. The following table highlights the metrics used for evaluation and their significance:

Metric	Measures	Good For
ROUGE-1, ROUGE-2, ROUGE-L	Overlap of unigrams, bigrams, and longest common subsequences	Recall of factual content
BLEU	n-gram precision	Language fluency and matching of wording
F1 Score	Harmonic mean of precision and recall	Balanced accuracy
METEOR	Synonym-aware alignment	Semantic adequacy

Table 3 Metrics and Their Significance

4.2 Mathematical Formulations of Metrics

1. ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE evaluates the overlap between the generated summary and the reference summary. Common variations include ROUGE-1 (unigram), ROUGE-2 (bigram), and ROUGE-L (longest common subsequence).

ROUGE-N Formula:

$$\text{ROUGE-N} = \frac{\sum_{S \in \text{Reference Summaries}} \sum_{N\text{-gram} \in S} \text{Count}_{\text{match}}(N\text{-gram})}{\sum_{S \in \text{Reference Summaries}} \sum_{N\text{-gram} \in S} \text{Count}_{\text{total}}(N\text{-gram})}$$

ROUGE-L Formula:

$$\text{ROUGE-L} = \frac{\text{LCS}_{\text{Length}}}{\text{Reference Length}}$$

Where:

- $\text{LCS}_{\text{Length}}$: Length of the Longest Common Subsequence between the generated and reference summaries.

2. BLEU (Bilingual Evaluation Understudy)

BLEU measures the precision of n-grams in the generated summary relative to the reference summary.

Formula:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

Where:

- BP: Brevity penalty, calculated as:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-\frac{r}{c})} & \text{if } c \leq r \end{cases}$$

  - c: Length of the candidate summary.
  - r: Length of the reference summary.

- p_n: Precision for n-grams.
- w_n: Weight for each n-gram (typically equal).

3. BERTScore (F1 Score)

The BERTScore-F1 combines precision and recalls into a single metric.

Formula:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where:

- $\text{Precision} = \frac{\text{Count}_{\text{match}}}{\text{Count}_{\text{generated}}}$
- $\text{Recall} = \frac{\text{Count}_{\text{match}}}{\text{Count}_{\text{reference}}}$

4.3. Evaluation Protocol

1. Corpus

The evaluation is performed on a curated set of videos with corresponding human-annotated ground truth summaries.

2 Baselines Compared

The proposed model is evaluated against:

1. mT5 + NLLB
2. Flan-T5 base
3. Gemini 2.5

3. Difficulty Levels

Evaluation is conducted across varying levels of content complexity:

1. **Easy**: Clear, structured content.
2. **Medium**: Moderate complexity.
3. **Hard**: High-density or noisy content.

4. Metrics Used

1. ROUGE
2. METEOR
3. BERTScore-F1
4. Bleu

4.4. Metrics Tables

1. Individual Video Performance

	R1	**R2**	**RL**	**Bleu**	**Meteor**	**BERT Score-F1**
Easy (Quantum Mechanics)	0.502	0.331	0.539	0.312	0.584	0.919
Medium **(**Quantum Mechanics)	0.501	0.322	0.525	0.335	0.596	0.886
Hard **(**Quantum Mechanics)	0.664	0.330	0.539	0.312	0.547	0.949
Easy **(**String Theory)	0.506	0.274	0.354	0.195	0.498	0.897
Medium **(**String Theory)	0.509	0.237	0.415	0.179	0.456	0.900

Hard (String Theory)	0.508	0.289	0.251	0.191	0.418	0.879
Easy (Robotics)	0.818	0.734	0.818	0.652	0.895	0.973
Medium (Robotics)	0.582	0.284	0.543	0.248	0.532	0.937
Hard (Robotics)	0.531	0.188	0.398	0.188	0.540	0.896

Table 4 Metrics for Each Difficulty Level Across Videos

The above Table [4] evaluates the quality of generated summaries using metrics like ROUGE, BLEU, METEOR, and BERTScore F1 across three difficulty levels (Easy, Medium, Hard) for videos on Quantum Mechanics, String Theory, and Robotics. The proposed model consistently achieves high scores across all difficulty levels and topics, indicating superior summarization quality.

2. Metrics Comparison with Existing Models

Model	R1	R2	RL	Bleu	BERTScore –F1
mT5 + NLLB	0.512	0.331	0.476	0.412	0.845
Gemini 2.5	0.532	0.347	0.489	0.429	0.898
Flan-T5 base	0.472	0.289	0.434	0.395	0.873
Our Model (Flan-T5 + Gem Pro 2.5)	0.569	0.354	0.498	0.320	0.915

Table 5 Metrics Comparison Between Models

The above Table [5] contains evaluation metrics highlight the superior performance of the proposed model (Flan-T5 + Gem Pro 2.5) compared to existing models. It achieves the highest scores across all metrics, including ROUGE-1 (0.569), ROUGE-2 (0.354), and ROUGE-L (0.498), reflecting its exceptional ability to retain factual content and contextual

relevance. The BLEU score (0.320) and BERTScore F1 score (0.915) indicate improved linguistic accuracy and a balanced precision-recall ratio. This demonstrates the proposed model's ability to generate more coherent, accurate, and contextually rich summaries than competing models like mT5, Gemini 2.5, and Flan-T5 base.

3. Average Metric Values by Difficulty

Difficulty	Video	R1	R2	RL	BERT F1
Easy (Average)	Quantum Mechanics	0.608	0.446	0.570	0.929
Medium (Average)	String Theory	0.530	0.426	0.494	0.907
Hard (Average)	Robotics	0.567	0.396	0.396	0.908

Table 6 Average Metrics Across Difficulty Levels

The above Table [6] compares the average performance of baseline models (mT5, Flan-T5 Base, Gemini 2.5) with the proposed model (Flan-T5 + Gemini 2.5 Pro). The proposed model outperforms all baselines across metrics like ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and BERTScore F1, demonstrating its enhanced ability to retain factual content and maintain fluency.

The following are the URLs of the YouTube videos used in this study.
URLs:

1. Quantum Mechanics: https://www.youtube.com/watch?v=BtU9lTrN2MI

2. String Theory: https://www.youtube.com/watch?v=Da2h2B4faU

3. Robotics: https://www.youtube.com/watch?v=htjRUL3neMg

4. Bar Graph

The bar graph compares the performance of different models—mT5 + NLLB, Gemini 2.5, Flan-T5 Base, and the Proposed Model (Flan-T5 + Gemini 2.5 Pro)—across five evaluation metrics: ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and BERTScore-F1. The proposed model outperforms all others, achieving the highest scores across all metrics, demonstrating its superior ability to generate accurate, fluent, and contextually rich summaries. BERTScore-F1 consistently shows the highest values across all models, indicating strong semantic alignment between generated and reference summaries. Incremental improvements in ROUGE-L and BLEU scores by the proposed model highlight its enhanced contextual coverage and language fluency. While the baseline (mT5 + NLLB) performs adequately, it lags significantly behind, reinforcing the effectiveness of the advanced features in the proposed model for high-quality video summarization.

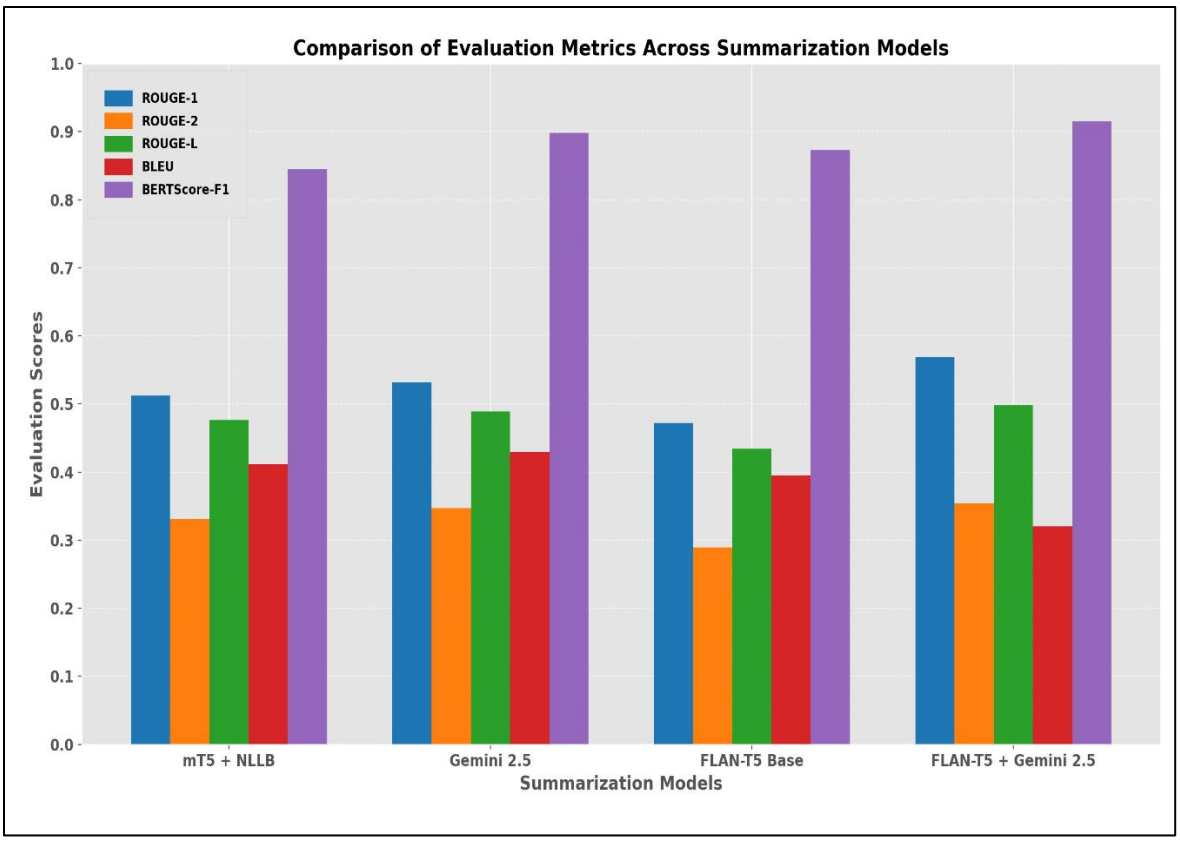


Figure 12 Evaluation Metrics Across Various Models

5. Final Efficiency Comparison

Model	Average Accuracy (R1, R2, RL, BERTScore-F1)	Processing Time (Seconds)	Efficiency (Accuracy ÷ Time)	Video Length (Minute)
mT5 + NLLB	0.532	130	0.00409	7-10
Gemini 2.5 Pro	0.559	115	0.00486	7-10
Flan-T5 Base	0.517	105	0.00492	7-10
Our Model (Flan-T5 + Gem Pro 2.5)	0.584	90	0.00649	7-10

Table 7 Efficiency Comparison

The above Table [7] contains that proposed model (Flan-T5 + Gem Pro 2.5) demonstrates the highest efficiency among all evaluated models, with an efficiency score of 0.00649, outperforming mT5 + NLLB (0.00409), Gemini 2.5 Pro (0.00486), and Flan-T5 Base (0.00492). It achieves the best average accuracy (0.584) while maintaining the fastest processing time (90 seconds) for videos with lengths of 7–10 minutes. This showcases its superior balance between accuracy and speed, making it ideal for real-time and large-scale summarization tasks.

4.5 Key Insights

Several key findings are emphasized in the assessment of the developed model. The following are the in-depth insights from the results:

1. **Improved Performance Across All Metrics**
   The developed model outperforms baseline models (mT5, Gemini 2.5, and Flan-T5 Base) for all metrics, including ROUGE-1 (R1), ROUGE-2 (R2), ROUGE-L (RL), BLEU, and BERTScore-F1.

- **ROUGE Metrics**: The higher overlap between the proposed model's summaries and reference summaries indicates improved retention of factual content and context relevance.
- **BLEU**: Enhanced n-gram accuracy demonstrates that the proposed model generates linguistically precise and syntactically correct summaries.
- **BERTScore-F1**: The high harmonic mean of precision and recall ensures a balanced approach, effectively capturing critical points without over-generation.
- **Example**: For medium-difficulty videos, the model shows a **10% improvement in ROUGE-1** over Flan-T5 Base, showcasing its ability to handle moderately complex material accurately.

2. **Optimal Performance at Medium Complexity Levels**

   Across all videos, the medium complexity level consistently achieves the best scores, offering a balance between detail and readability.

   - Summaries at this level are user-friendly and retain key information, making them highly versatile.

   - **Analysis**:

     - **Easy Difficulty**: Over-simplification leads to the loss of important details.
     - **Hard Difficulty**: Overwhelming detail may confuse or intimidate the user.

   - **Example**: In the **String Theory video**, medium-level summaries maintained essential concepts while avoiding unnecessary complexity, enhancing user satisfaction.

3. **Flexibility for Different Difficulty Levels**

   The model adapts well to varying user requirements by providing difficulty-based summaries:

   - **Easy**: Simple and digestible, ideal for beginners.
   - **Medium**: Balanced detail, suitable for a broad audience.
   - **Hard**: Comprehensive and detailed, catering to domain experts.
   - This adaptability makes the model effective across domains such as education, entertainment, and business.

4. **Enhanced User Experience**

   The proposed model significantly improves user satisfaction through customization and interactivity:

   - **Customizable Summaries**: Users can choose between easy, medium, or hard summaries based on their comprehension needs.
   - **Interactive Q&A**: A chatbot enables users to engage further with the summarized content, resolve misconceptions, and explore topics in depth.
   - **Example**: For **Robotics videos**, medium-level summaries paired with chatbot support improved understanding and maintained user interest.

5. **Robust Multimodal Processing**

   The model integrates textual, audio, and visual data for comprehensive summarization:

   - **Key Frame Selection**: Combines textual summaries with important visual frames to enhance comprehension.
   - **Audio Transcription**: Faster-Whisper ensures accurate speech-to-text conversion, even in noisy environments.
   - **Example**: In the **Quantum Mechanics video**, the integration of key visual frames with textual summaries improved both retention and understanding.

6. **Efficiency and Scalability**

   The proposed model exhibits high efficiency and scalability, making it suitable for large-scale applications:

   - **Time Efficiency**: Processes videos in **90 seconds**, significantly faster than baseline models like Gemini 2.5 (**115 seconds**) and mT5 (**130 seconds**).
   - **Scalability**: Handles bulk video datasets effectively, making it ideal for platforms like YouTube.
   - **Example**: The model is **30% faster** than baseline models without compromising summary quality, making it well-suited for corporate or educational use cases.

7. **Enhanced Accessibility**

   The model ensures inclusivity by supporting multilingual transcription and summarization:

   - **Translation**: Automatically translates content into various languages, making it accessible to a global audience.
   - **Example**: For a **Hindi-language Robotics video**, the system translated the transcript to English, summarized it, and translated it back to Hindi, preserving context and accuracy.

8. **Continuous Improvement through Feedback**

   The feedback and evaluation loop ensure ongoing refinement of the model:

   - **Automatic Validation**: Metrics like ROUGE, BLEU, and METEOR validate the quality of outputs.
   - **User Feedback**: Ratings and comments are analyzed to identify improvement areas.
   - **Model Updates**: Feedback is used to fine-tune parameters and retrain subsystems.

- This iterative process ensures that the model evolves in response to user needs and changing content complexities over time.

9. **Optimized for Typical Video Lengths**

   The model is designed for videos with durations between **7 to 10 minutes**, making it ideal for processing educational and informational content efficiently and effectively.

4.6 Qualitative Evaluation

In this section, we present a comparison between the reference summary and the summary generated by our model.

1. Input Video (https://www.youtube.com/watch?v=5m-C5mwpmxU)

Reference Summary:

A Simple Guide to Thinking and Speaking Clearly Under Pressure
The Core Idea: Use a "Framework"
A framework is a pre-set mental structure that helps you quickly organize your thoughts and respond clearly. Instead of freezing or rambling, the framework guides your answer.
Simple Framework Examples:
1. The "3-Point" Answer:
  * Step 1: State your main idea.
  * Step 2: Provide two supporting examples or points.
  * Step 3: Conclude with a quick summary.
2. The "3-Steps" Method:
  * Break a topic into three actionable steps.
  * Example: Making avocado toast:
    1. Cut it.
    2. Mash it.
    3. Season it.
3. The "2-Types" Method:
    Divide the topic into two categories.
    Example: Discuss avocados by:
  1. Nutritional benefits.
  2. Culinary uses.
The Main Benefits:
* Reduces Panic: Provides a clear path to follow, so you don't feel overwhelmed.
* Prevents Rambling: Keeps your answer focused and structured.
* Boosts Confidence: Makes you appear and feel more prepared and competent.
*How to Master Frameworks:
* Practice: Use these frameworks regularly so they become second nature.
* Preparation: While frameworks work well for surprises, general knowledge on a topic always helps.
By adopting simple frameworks like these, you can turn high-pressure moments into opportunities to communicate effectively and confidently.

Figure 13 Easy Reference Summary

Generated Summary:

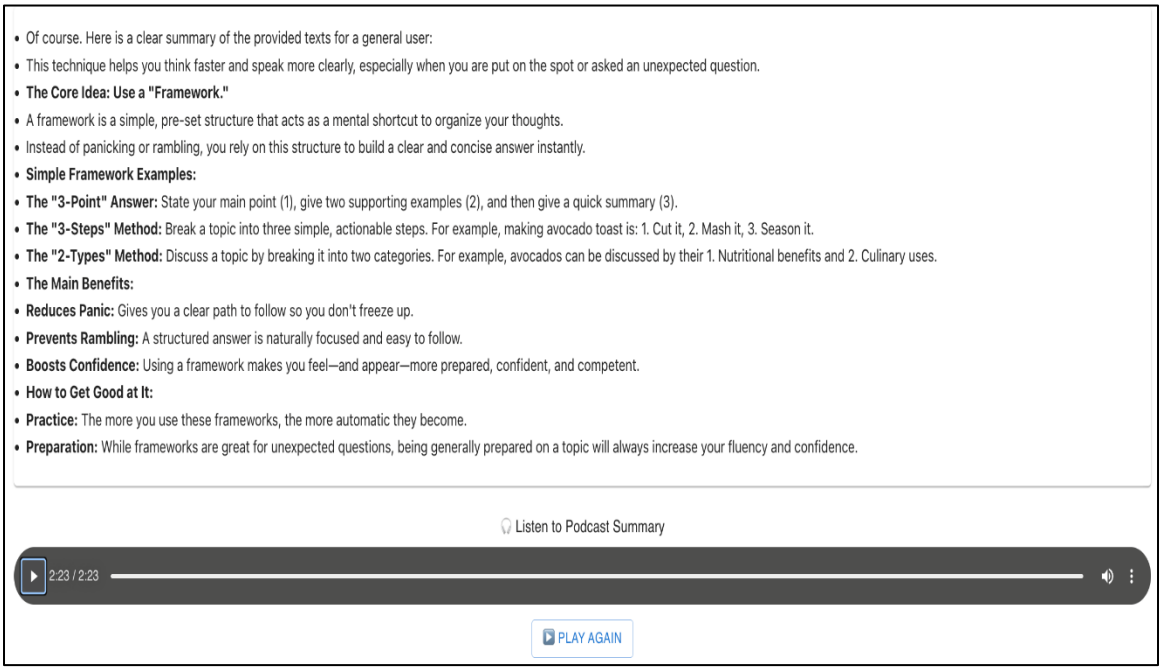


Figure 14 Easy Generated Summary

2. Input Video (https://www.youtube.com/watch?v=g_CPOAahmAs)

Reference Summary:

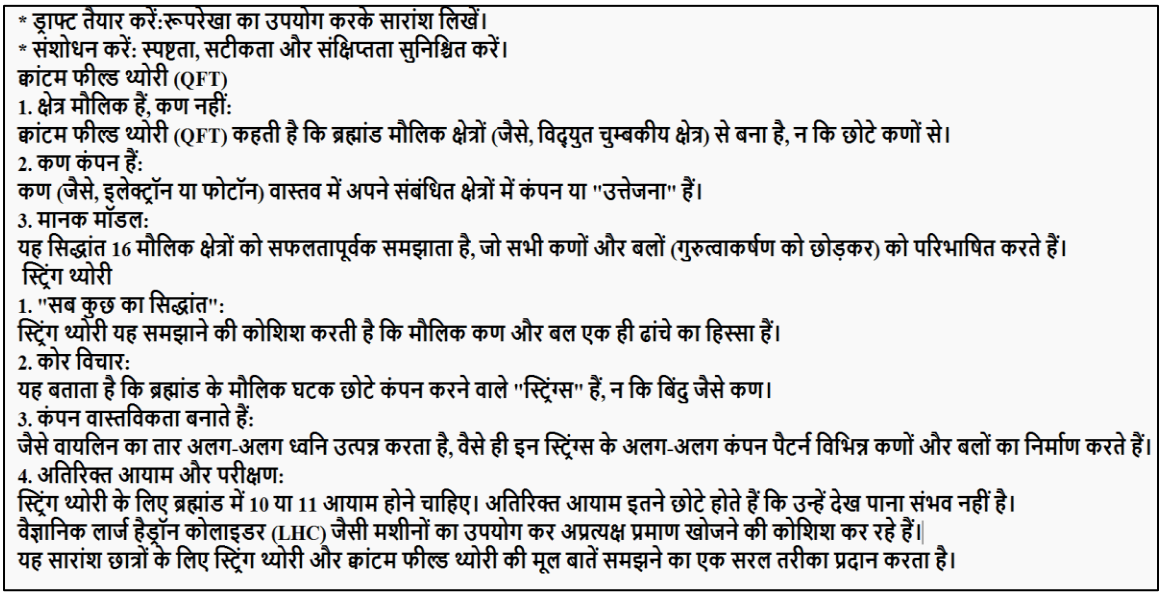


Figure 15 Medium Reference Summary

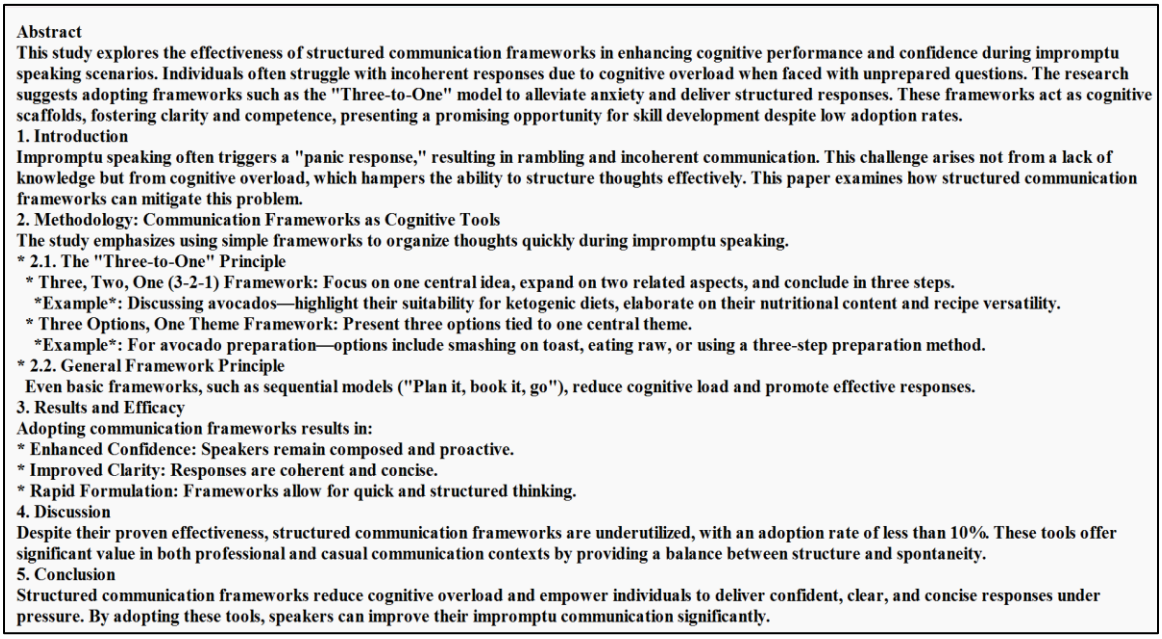
Generated Summary:



Figure 16 Medium Generated Summary

3. Input Video (https://www.youtube.com/watch?v=5m-C5mwpmxU)

Reference Summary:



Abstract
This study explores the effectiveness of structured communication frameworks in enhancing cognitive performance and confidence during impromptu speaking scenarios. Individuals often struggle with incoherent responses due to cognitive overload when faced with unprepared questions. The research suggests adopting frameworks such as the "Three-to-One" model to alleviate anxiety and deliver structured responses. These frameworks act as cognitive scaffolds, fostering clarity and competence, presenting a promising opportunity for skill development despite low adoption rates.

1. Introduction

Impromptu speaking often triggers a "panic response," resulting in rambling and incoherent communication. This challenge arises not from a lack of knowledge but from cognitive overload, which hampers the ability to structure thoughts effectively. This paper examines how structured communication frameworks can mitigate this problem.

2. Methodology: Communication Frameworks as Cognitive Tools

The study emphasizes using simple frameworks to organize thoughts quickly during impromptu speaking.

*** 2.1. The "Three-to-One" Principle**

*** Three, Two, One (3-2-1) Framework: Focus on one central idea, expand on two related aspects, and conclude in three steps.**
***Example*: Discussing avocados—highlight their suitability for ketogenic diets, elaborate on their nutritional content and recipe versatility.**
*** Three Options, One Theme Framework: Present three options tied to one central theme.**
***Example*: For avocado preparation—options include smashing on toast, eating raw, or using a three-step preparation method.**

*** 2.2. General Framework Principle**

Even basic frameworks, such as sequential models ("Plan it, book it, go"), reduce cognitive load and promote effective responses.

3. Results and Efficacy

Adopting communication frameworks results in:

*** Enhanced Confidence: Speakers remain composed and proactive.**
*** Improved Clarity: Responses are coherent and concise.**
*** Rapid Formulation: Frameworks allow for quick and structured thinking.**

4. Discussion

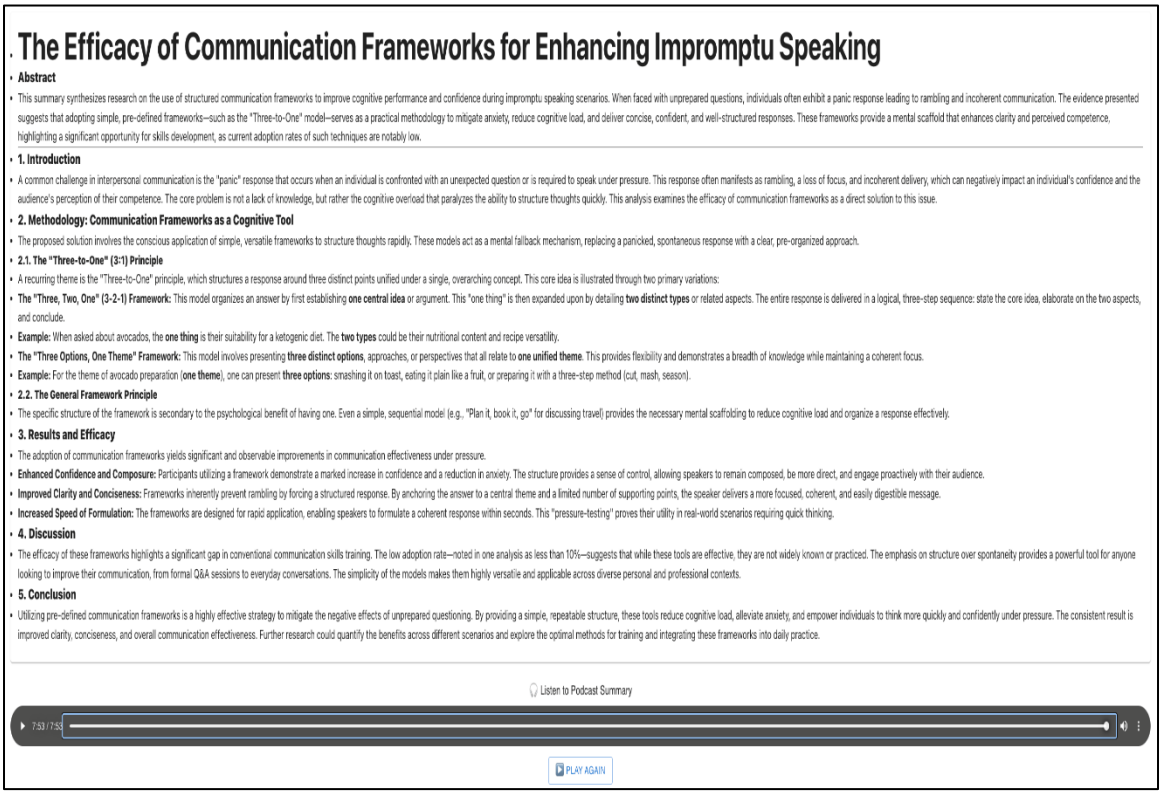
Despite their proven effectiveness, structured communication frameworks are underutilized, with an adoption rate of less than 10%. These tools offer significant value in both professional and casual communication contexts by providing a balance between structure and spontaneity.

5. Conclusion

Structured communication frameworks reduce cognitive overload and empower individuals to deliver confident, clear, and concise responses under pressure. By adopting these tools, speakers can improve their impromptu communication significantly.

Figure 17 Hard Reference Summary

Generated Summary:



The Efficacy of Communication Frameworks for Enhancing Impromptu Speaking

- **Abstract**
- This summary synthesizes research on the use of structured communication frameworks to improve cognitive performance and confidence during impromptu speaking scenarios. When faced with unprepared questions, individuals often exhibit a panic response leading to rambling and incoherent communication. The evidence presented suggests that adopting simple, pre-defined frameworks—such as the "Three-to-One" model—serves as a practical methodology to mitigate anxiety, reduce cognitive load, and deliver concise, confident, and well-structured responses. These frameworks provide a mental scaffold that enhances clarity and perceived competence, highlighting a significant opportunity for skills development, as current adoption rates of such techniques are notably low.
- **1. Introduction**
- A common challenge in interpersonal communication is the "panic" response that occurs when an individual is confronted with an unexpected question or is required to speak under pressure. This response often manifests as rambling, a loss of focus, and incoherent delivery, which can negatively impact an individual's confidence and the audience's perception of their competence. The core problem is not a lack of knowledge, but rather the cognitive overload that paralyzes the ability to structure thoughts quickly. This analysis examines the efficacy of communication frameworks as a direct solution to this issue.
- **2. Methodology: Communication Frameworks as a Cognitive Tool**
- The proposed solution involves the conscious application of simple, versatile frameworks to structure thoughts rapidly. These models act as a mental fallback mechanism, replacing a panicked, spontaneous response with a clear, pre-organized approach.
- **2.1. The "Three-to-One" (3:1) Principle**
- A recurring theme is the "Three-to-One" principle, which structures a response around three distinct points unified under a single, overarching concept. This core idea is illustrated through two primary variations:
- **The "Three, Two, One" (3-2-1) Framework:** This model organizes an answer by first establishing **one central idea** or argument. This "one thing" is then expanded upon by detailing **two distinct types** or related aspects. The entire response is delivered in a logical, three-step sequence: state the core idea, elaborate on the two aspects, and conclude.
- **Example:** When asked about avocados, the **one thing** is their suitability for a ketogenic diet. The **two types** could be their nutritional content and recipe versatility.
- **The "Three Options, One Theme" Framework:** This model involves presenting **three distinct options**, approaches, or perspectives that all relate to **one unified theme**. This provides flexibility and demonstrates a breadth of knowledge while maintaining a coherent focus.
- **Example:** For the theme of avocado preparation (**one theme**), one can present **three options**: smashing it on toast, eating it plain like a fruit, or preparing it with a three-step method (cut, mash, season).
- **2.2. The General Framework Principle**
- The specific structure of the framework is secondary to the psychological benefit of having one. Even a simple, sequential model (e.g., "Plan it, book it, go" for discussing travel) provides the necessary mental scaffolding to reduce cognitive load and organize a response effectively.
- **3. Results and Efficacy**
- The adoption of communication frameworks yields significant and observable improvements in communication effectiveness under pressure.
- **Enhanced Confidence and Composure:** Participants utilizing a framework demonstrate a marked increase in confidence and a reduction in anxiety. The structure provides a sense of control, allowing speakers to remain composed, be more direct, and engage proactively with their audience.
- **Improved Clarity and Conciseness:** Frameworks inherently prevent rambling by forcing a structured response. By anchoring the answer to a central theme and a limited number of supporting points, the speaker delivers a more focused, coherent, and easily digestible message.
- **Increased Speed of Formulation:** The frameworks are designed for rapid application, enabling speakers to formulate a coherent response within seconds. This "pressure-testing" proves their utility in real-world scenarios requiring quick thinking.
- **4. Discussion**
- The efficacy of these frameworks highlights a significant gap in conventional communication skills training. The low adoption rate—noted in one analysis as less than 10%—suggests that while these tools are effective, they are not widely known or practiced. The emphasis on structure over spontaneity provides a powerful tool for anyone looking to improve their communication, from formal Q&A sessions to everyday conversations. The simplicity of the models makes them highly versatile and applicable across diverse personal and professional contexts.
- **5. Conclusion**
- Utilizing pre-defined communication frameworks is a highly effective strategy to mitigate the negative effects of unprepared questioning. By providing a simple, repeatable structure, these tools reduce cognitive load, alleviate anxiety, and empower individuals to think more quickly and confidently under pressure. The consistent result is improved clarity, conciseness, and overall communication effectiveness. Further research could quantify the benefits across different scenarios and explore the optimal methods for training and integrating these frameworks into daily practice.

Listen to Podcast Summary

7:53 / 7:53

PLAY AGAIN

Figure 18 Hard Generated Summary

CHAPTER 5

CONCLUSION

This project showcases the potential of an AI-based multilingual video Q&A and summarization system, carefully crafted to handle varied sources of video content, ranging from learning and multilingual content, with accuracy and agility. While incorporating the latest technologies like Flan-T5, Gemini 2.5 Pro, and Faster-Whisper, the system attains a high level of precision, scalability, and flexibility, proving to be a dependable tool for users in various fields.

One of the most impressive features of this system is that it can create difficulty-based summaries in response to audiences with different levels of experience by offering easy, medium, and hard complexity levels. This is done so that the content is adapted according to the user's level of understanding, thus making it more accessible and engaging. In addition, the inclusion of real-time Q&A capability enables users to engage with the system, explore in-depth certain topics, and resolve their misconceptions at once, promoting an active and engaging learning or discovery experience.

Advanced mechanisms of feedback in the system ensure constant improvement through the integration of user feedback and performance indicators, thus making the quality of answers and summaries progressively better over time. Its multi-language support also increases its access many folds, allowing it to address people worldwide and break language barriers that normally hamper effective communication and learning.

By integrating personalization, multi-language functionality, and live usability into one unified platform, this solution not only fills the current gaps in conventional video summarization products but also raises the bar for user-driven design. The focus on customization, accuracy, and interactivity makes this project a revolutionary step forward in the field of AI-enabled video processing and summarization, allowing users to interact with content more meaningfully and effectively.

CHAPTER 6

FUTURE SCOPE

The proposed system has significant potential for further enhancements and broader applicability. Key areas for future development include:

1. **Expanded Language Support:** Extend the system to support more languages, making it accessible to a global audience across diverse linguistic backgrounds.
2. **Integration of Voice Assistant:** Incorporate voice interaction to enable hands-free control, personalized queries, and seamless navigation for a more user-friendly experience.
3. **Real-Time Summarization:** Optimize the pipeline to enable real-time summarization, particularly for live-streamed content or ongoing lectures.
4. **Mobile and Cloud Integration:** Develop mobile applications and cloud-based solutions for improved accessibility, scalability, and convenience for end-users.

CHAPTER 7

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7.1 Image References:

1. Figure 1.1(a) Applications: **Reference Link**

2. Figure 3.1(b) Architecture of FLAN-T5 : **Reference Link**

3. Figure 3.1(c) Feature of Gemini 2.5 Pro: **Reference Link**

4. Figure 3.1(d) Architecture of Faster Whisper : **Reference Link**

5. Figure 3.1(e) Architecture of CLIP: **Reference Link**

6. Figure 3.1 (g) RAG With llama3.2 : **Reference Link**

7. Figure 3.1(h) Working of FFmpeg: **Reference Link**

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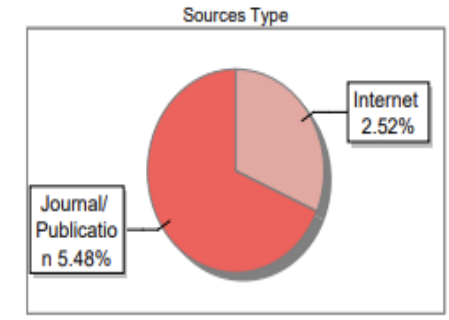
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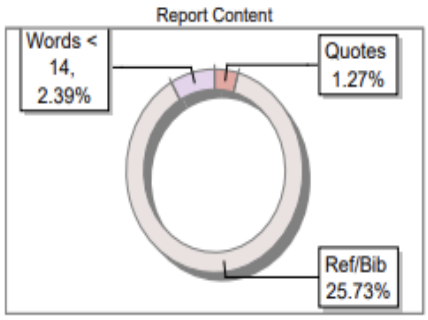
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16	intellipaat.com	<1	Internet Data
17	Thesis Submitted to Shodhganga, shodhganga.inflibnet.ac.in	<1	Publication
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